

MAPPING OF ALIENABLE LAND IN BHUTAN



Technical Report 2: Desktop Verification and Field Validation



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This report was prepared under the National Land Use Zoning (NLUZ) initiative coordinated by the National Land Commission Secretariat (NLCS) and is a sub-report of Technical Report 1 and other project credits remain inline with the Technical Report 1

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EXECUTIVE SUMMARY

This report presents the desktop verification and field validation of high-suitability land parcels identified under the Alineable Land Inventory initiative. It builds upon the outputs of Report 1, which established a national baseline of alienable land using GIS-based Multi-Criteria Decision Analysis (MCDA).

Focusing on the highest suitability areas, with minor refinements to classification criteria (Appendix-I), this study applies a two-stage validation approach. Desktop verification was first conducted using high-resolution satellite imagery and supporting topographic base maps to reassess and confirm model-derived outputs through spatial and visual analysis. Field validation was subsequently carried out at selected sites across six dzongkhags to ground-truth the results and verify on-ground conditions.

The results indicate that the initial model outputs are generally consistent with observed conditions, with desktop verification significantly improving spatial accuracy through geometric consolidation and boundary optimization. Field validation further strengthens confidence in the refined inventory while also identifying site-specific constraints.

However, certain limitations remain, including terrain-related constraints, environmental considerations such as overlaps with managed forest regimes, and data gaps related to localized hazards.

Overall, this report provides a refined and validated inventory of high-suitability land parcels, supporting evidence-based planning, investment prioritization, and policy decision-making under the NLUZ initiative. The dataset also serves as a foundational input for the development of the Land Bank Management System (LBMS).

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Acronyms

Acronym	Full Form
MCDA	Multi-Criteria Decision Analysis
GIS	Geographic Information System
NLUZ	National Land Use Zoning
NLCS	National Land Commission Secretariat
DoFPS	Department of Forest and Park Services
NSSC	National Soil Service Centre
WWF	World Wide Fund for Nature
DEM	Digital Elevation Model
TCC	Tree Crown Cover
LULC	Land Use / Land Cover
RoW	Right of Way
CF	Community Forest
FMU	Forest Management Unit
HCV	High Conservation Value
KBA	Key Biodiversity Area
SFM	Sustainable Forest Management
LBMS	Land Bank Management System

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1. INTRODUCTION

Following the national-level suitability modeling in Technical Report 1, this second phase shifts from a broad assessment of potential to a targeted operational inventory. The initial model identified a wide extent of “Highly Suitable” (S1) land derived from the combined Class 4 and Class 5 suitability tiers (as defined in Appendix I of Technical Report 1).

To ensure a practical and high-confidence starting point for validation, the classification criteria were refined through strategic re-parameterization. While Class 4 areas were theoretically suitable, the lower portions of this class were found to involve ground-level complexities that did not consistently meet the threshold for very high suitability.

As a result, this report defines a refined “Search Area” consisting of Class 5 parcels and the high-confidence upper portion of Class 4, consolidated into a single priority tier for verification. The re-parameterization process is described in Appendix I of this report. This refinement ensures that desktop and field validation efforts are focused on parcels with the highest likelihood of immediate availability and technical readiness.

2. METHODOLOGY OVERVIEW

This study adopts a validation-focused approach building upon the MCDA framework developed in Report 1. The methodology consists of three key stages:

1. Extraction of High-Suitability Areas

High-suitability (Class 5) parcels identified in Report 1 were extracted as the primary input dataset.

2. Desktop Verification

Each parcel was evaluated using high-resolution satellite imagery and supporting spatial datasets. Parcels were categorized into feasibility classes based on observable characteristics such as proximity to rivers, settlement patterns, access, slope conditions, and land cover.

3. Field Validation

Selected sites across six dzongkhags were visited to validate desktop observations. Field teams assessed terrain, accessibility, land use, and development constraints.

The process ensures that model-derived outputs are refined into practically usable and policy-relevant land information.

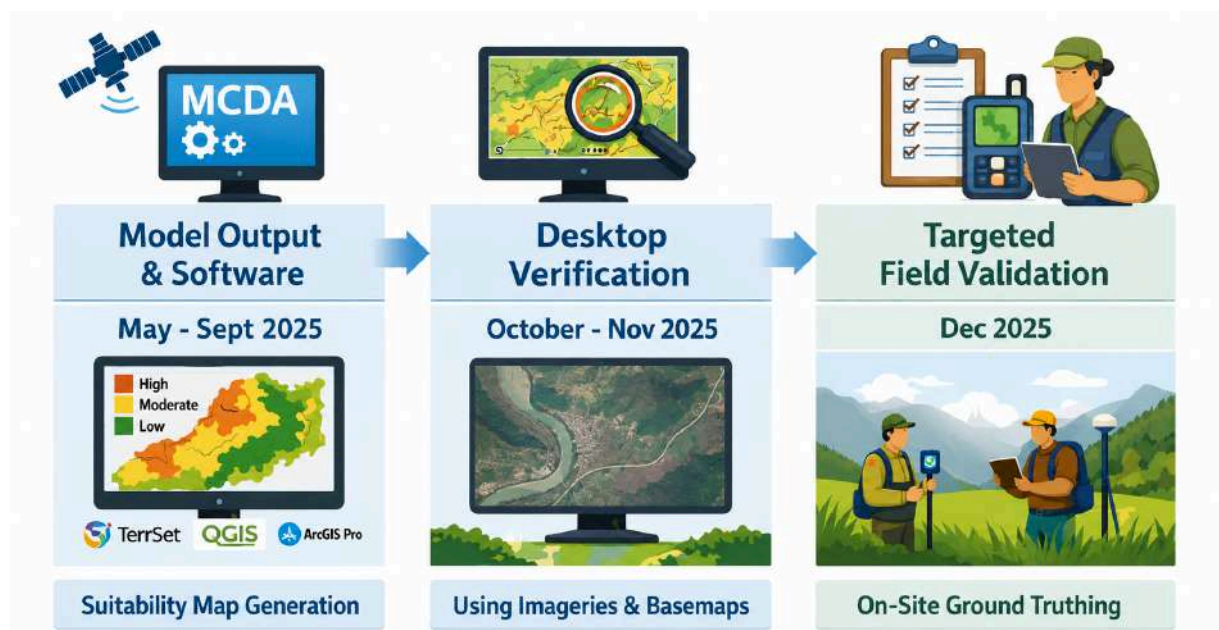


Figure 1: Timeline of the Alienable Land Identification and Verification Process

3. DATA PREPARATION FOR DESKTOP VERIFICATION

The preparation of high-potential land parcels for desktop verification involved three main steps: generation of high-potential areas, application of constraints, and data refinement and cleaning. The complete step-by-step procedure, including factor re-parameterization, workflow operations, tables, and figures, is provided in Annexure 1.

3.1 Generation of High-Potential Areas

High-suitability land parcels were identified using a Multi-Criteria Decision Analysis (MCDA) conducted in TerrSet. The analysis incorporated six key factors: slope, settlement proximity, water sources, road proximity, Tree Crown Cover (TCC), and elevation.

Each factor was re-parameterized to suit local conditions, with suitability classes ranging from 1 (Very Low/Unsuitable) to 5 (Very High Suitability). The MCDA produced a raster output, from which Class 5 areas, representing highly suitable land, were extracted and vectorized in QGIS for further processing.

Detailed factor weight tables, suitability thresholds, and raster-to-vector conversion steps are provided in Annexure 1.

3.2 Application of Constraints

To ensure that the identified high-potential areas are realistic and implementable, several spatial constraints were applied:

- **Option 3 restricted areas** were erased from the dataset. Option 3 includes unsuitable land identified from land use/land cover (LULC), cadastral restrictions, and core zones of protected areas (see **Technical Report 1**).
- **Powerline buffer removal** eliminated areas intersecting powerline corridors.
- **Highway Right of Way (RoW) removal** excluded areas near highways.

Step-by-step workflows, including QGIS operations and pre- and post-removal illustrations, are included in **Appendix-I**.

3.3 Data Refinement and Cleaning

The resultant Class 5 dataset was further refined to ensure accuracy and usability for desktop verification:

- **Polygon generalization:** Sharp edges of polygons were smoothed using buffer and dissolve operations.
- **Removal of small areas:** Polygons smaller than 1 acre were excluded from the dataset.
- **Automated cleaning:** Geometry fixes, multipart-to-singlepart conversion, and recalculation of areas in QGIS were carried out to ensure spatial integrity.

The final dataset represents high-potential land parcels ready for desktop verification, free from irrelevant overlaps and spatial inconsistencies. Detailed procedures for generalization, area recalculation, and cleaning operations are documented in Appendix-I.

3.4 Summary of Data Preparation

This workflow ensured that Class 5 high-potential areas were systematically generated, relevant constraints applied, and the data refined to a clean and accurate state. The resultant dataset is suitable for **desktop verification** and further analysis.

3.5 Analysis of Data Selected for Desktop Verification

3.5.1 Rationale for Selection

Factor re-parameterization was carried out to identify the most suitable areas for desktop verification, with the aim of improving the reliability and confidence of the data.

This step was necessary because desktop validation is a manual and time-intensive process, and verifying the entire Part-I dataset would be a mammoth task. Therefore, the focus was narrowed to improve accuracy and increase the probability of land availability.

3.5.2 Selection Approach

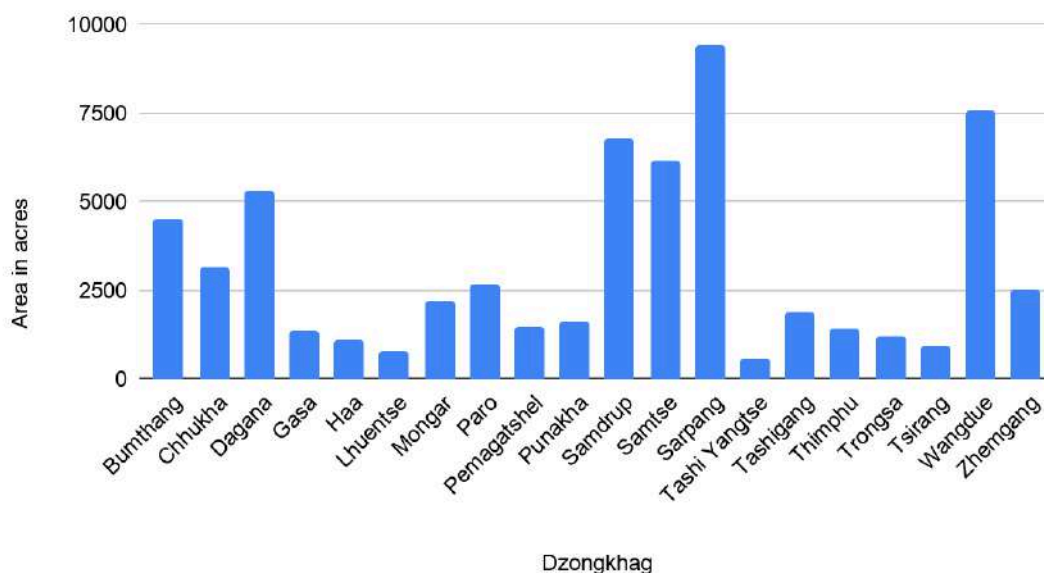
The parameters were selected to determine **potentially available alienable land**. However, for specific land-use planning, suitability analysis (GIS-based MCDA) may be carried out using additional or modified parameters as required.

3.6 Dzongkhag-wise Distribution of Identified Land

Table 1: Dzongkhag-wise statistics of highly suitable areas before desktop verification

<i>Dzongkhag</i>	Area in acres
Bumthang	4524.358
Chhukha	3145.258
Dagana	5317.766
Gasa	1376.689
Haa	1110.158
Lhuentse	794.251
Mongar	2224.705
Paro	2661.292
Pemagatshel	1463.811
Punakha	1641.298
Samdrup Jongkhar	6780
Samtse	6139.027
Sarpang	9393.722
Tashi Yangtse	546.375
Tashigang	1888.168
Thimphu	1393.125
Trongsa	1194.846
Tsirang	932.288
Wangdue Phodrang	7565.262
Zhemgang	2542.797
Grand Total	62635.196

Figure 2: Statistics of Highly Suitable Areas Prior to Desktop Verification



Analysis of Identified Highly Suitable Alienable Land

The software-based analysis identified a total of 62,635.196 acres of highly suitable alienable land across all Dzongkhags. The distribution indicates notable spatial variation, with Sarpang (9,393.722 acres), Wangdue Phodrang (7,565.262 acres), Samdrup Jongkhar (6,780 acres), and Samtse (6,139.027 acres) accounting for the largest shares of suitable land. In contrast, Dzongkhags such as Tashi Yangtse (546.375 acres), Lhuentse (794.251 acres), and Haa (1,110.158 acres) exhibit relatively limited extents of highly suitable areas, reflecting terrain constraints and existing land use patterns.

Following the preparation and refinement of high-suitability parcels, desktop verification was conducted as the first stage of validation to assess practical feasibility prior to field assessment.

4. DESKTOP VERIFICATION APPROACH

The above output represents the initial identification of highly suitable alienable land generated through GIS-based spatial analysis. Following this step, the identified areas were further verified using base datasets and open satellite imagery to assess on-ground conditions, validate land cover and accessibility, and minimize potential misclassification. This approach ensures that the results are not solely model-driven but are refined through desktop verification prior to any field-based assessment.

Desktop verification was undertaken through systematic manual validation of all parcels classified as highly suitable. The process utilized multiple authoritative datasets, including 1:25,000 topographic base maps, CartoSAT DEM, DEM derived from topographic maps for slope analysis, cadastral geodatabase, road network data, and open-source satellite imagery. Parcels identified as unsuitable through visual interpretation, supported by these ancillary datasets, were removed from the inventory, while parcels retaining potential suitability were preserved and further classified into four broad categories to support structured assessment and subsequent analysis.

1. **OK** – Parcels that appear suitable based on desktop verification with no obvious constraints.
2. **Near to River** – Parcels that appear suitable but are located in proximity to rivers. Due to limited availability of river hazard and flood-related data, associated risks cannot be ruled out; therefore, field validation by relevant domain experts is required.
3. **Existing Structure** – Parcels where existing structures are visible in satellite imagery (Google Earth, Bing), but corresponding map updates are absent in the cadastral geodatabase. These cases require field verification to ascertain whether they involve pending cadastral updates or unauthorized occupation.
4. **Cultivated** – Parcels that appear to be under agricultural use based on satellite imagery, with missing or outdated cadastral records. Field validation is necessary to confirm current land use and legal status.

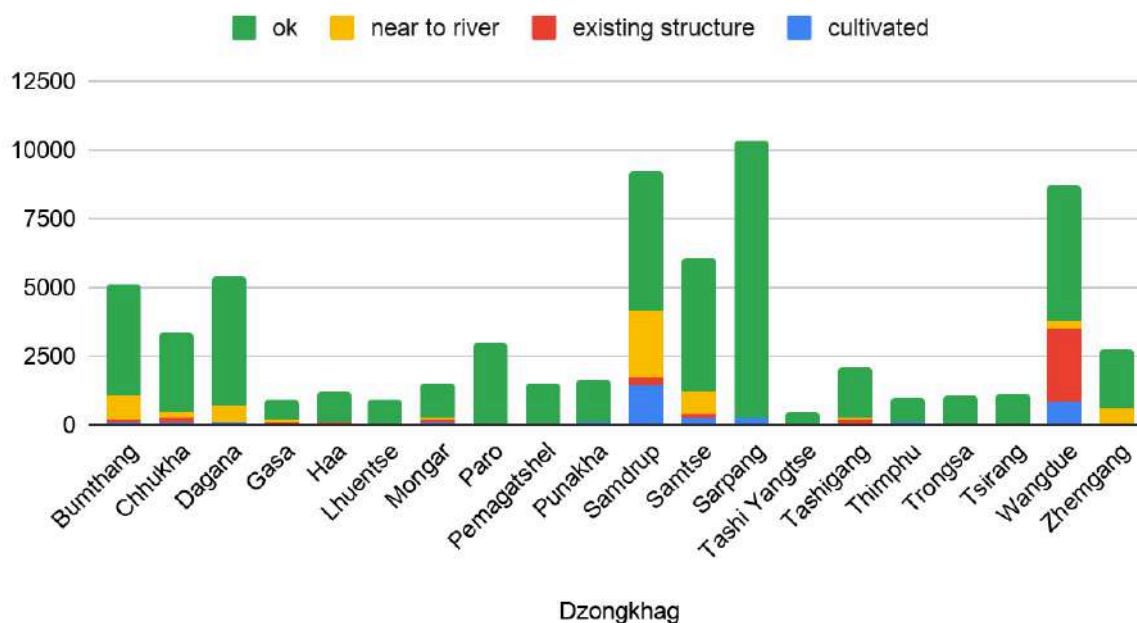
This structured desktop verification framework ensures that the subsequent statistical analysis reflects areas with a higher probability of availability, while clearly identifying parcels that require further field validation or institutional consultation.

4.1 Area by Dzongkhag and Desktop Verification Status

Table 2: Dzongkhag-wise statistics of highly suitable areas after desktop verification

<i>Dzongkhag</i>	cultivated	existing structure	near to river	ok	Grand Total
Bumthang	79.906	77.781	892.344	4053.861	5103.892
Chhukha	108.761	125.017	226.776	2878.13	3338.684
Dagana	87.678	52.232	555.647	4734.008	5429.565
Gasa	1.682	73.996	66.705	755.924	898.307
Haa	12.283	74.28		1099.435	1185.998
Lhuentse	6.755	27.234	2.997	870.561	907.547
Mongar	107.589	99.263	22.587	1279.5	1508.939
Paro	6.051	53.113		2894.23	2953.394
Pemagatshel		28.5	12.399	1429.583	1470.482
Punakha	67.864	43.382	1.868	1522.907	1636.021
Samdrup Jongkhar	1414.923	289.63	2470.339	5034.586	9209.478
Samtse	234.775	172.928	784.424	4869.769	6061.896
Sarpang	223.294	41.237		10072.746	10337.277
Tashi Yangtse	8.233	15.362		451.127	474.722
Tashigang	7.622	189.619	47.806	1828.807	2073.854
Thimphu	71.569	38.496		900.943	1011.008
Trongsa	1.741	3.192	15.392	1029.442	1049.767
Tsirang	19.941	41.499		1098.06	1159.5
Wangdue Phodrang	874.26	2625.874	282.081	4883.104	8665.319
Zhemgang	24.615	17.962	540.743	2156.664	2739.984
Grand Total	3359.542	4090.597	5922.108	53843.387	67215.634

Figure 3: Desktop Verification Status by Dzongkhag



Analysis of Desktop-Verified Parcels

The desktop verification resulted in a total of 67,215.634 acres of parcels retained in the inventory after initial screening. Of this, the majority—53,843.387 acres (approximately 80%)—fall under the “OK” category, indicating parcels that appear suitable with no major constraints identified through desktop verification. This reinforces the effectiveness of the re-parameterization approach in narrowing the analysis toward areas with a higher probability of availability.

Parcels classified as “Near to River” account for 5,922.108 acres, representing a significant secondary category. Dzongkhags such as Samdrup Jongkhar, Dagana, Wangdue Phodrang, and Zhemgang show notable extents in this class, reflecting the influence of river systems on land suitability. While these parcels appear suitable, their proximity to rivers necessitates field-level hazard assessment, particularly due to the limited availability of flood and river hazard datasets at the national scale.

The “Existing Structure” category comprises 4,090.597 acres, with higher concentrations observed in Wangdue Phodrang, Samdrup Jongkhar, and Tashigang. These cases point to potential gaps between on-ground development and cadastral updates, highlighting the importance of field verification to determine whether such parcels involve pending regularization or unauthorized occupation.

The “Cultivated” category accounts for 3,359.542 acres. Higher values in Dzongkhags such as Samdrup Jongkhar, Wangdue Phodrang, and Samtse suggest the presence of active agricultural use that is either missing or not fully reflected in the cadastral geodatabase. These parcels require careful field validation to avoid conflicts with existing livelihoods and land-use practices.

At the Dzongkhag level, Sarpang, Samdrup Jongkhar, Wangdue Phodrang, and Samtse collectively contribute a substantial share of the total verified inventory, indicating relatively higher potential availability of alienable land in these Dzongkhags. In contrast, Dzongkhags such as Gasa, Haa, Tashi Yangtse, and Trongsa show comparatively lower extents, largely reflecting terrain constraints, conservation considerations, and existing land-use patterns.

Overall, the statistical distribution demonstrates that while a significant proportion of parcels exhibit high suitability, a non-negligible share falls under categories that require field validation and institutional consultation. This classification-based statistical breakdown provides a clear basis for prioritizing field verification efforts, focusing resources on parcels with higher uncertainty while maintaining confidence in areas classified as “OK”.

4.2 Assessment of Alienable Land Within Sustainable Forest Management Regimes

Prior to desktop verification, parcels located within the core and transition zones of protected areas and biological corridors were excluded from the analysis. However, Key Biodiversity Areas (KBA) and High Conservation Value (HCV) areas were not excluded at this stage.

During desktop verification, overlay analysis was conducted to identify parcels overlapping with KBA, HCV, Forest Management Units (FMU), and Community Forests (CF). These areas were retained in the inventory, as a substantial portion of the identified alienable land lies within such boundaries. Retaining these parcels allows for evidence-based decision-making, whereby any land release, if considered, would be subject to consultation with the relevant authorities.

Table 3: Area overlap statistics with forest management regimes

<i>lu_overlap</i>	area in acres
CF	8758.249
CF,FMU	653.343
CF,FMU,HCV-KBA	13.861
CF,HCV-KBA	166.951
FMU	4610.1019
FMU,HCV-KBA	35.177
HCV-KBA	194.802
Grand Total	14432.4849

The overlay analysis shows that 14,432.485 acres of the desktop-verified alienable land inventory overlap with existing forest management and conservation-related boundaries. The largest proportion of overlap occurs within Community Forests (CF), accounting for 8,758.249 acres, followed by Forest Management Units (FMU) with 4,610.102 acres. Together, CF and FMU-related overlaps constitute the majority of the total, indicating that a significant share of potentially alienable land lies within managed forest landscapes.

Areas with combined overlaps such as CF and FMU (653.343 acres) and overlaps involving HCV-KBA are comparatively limited. Parcels overlapping HCV-KBA alone (194.802 acres) and in combination with CF or FMU collectively represent a relatively small fraction of the total overlap area.

These overlapping parcels were not excluded during desktop verification, as doing so would have substantially reduced the inventory and constrained planning options. Instead, they were deliberately retained to allow case-by-case evaluation, where any consideration of land release would require consultation with relevant forest and conservation authorities and compliance with applicable regulations.

Overall, the overlap analysis provides a transparent risk-screening layer, clearly identifying parcels that may require institutional coordination while ensuring that decision-making remains flexible, informed, and evidence-based.

4.3 Comparison of Pre- and Post-Desktop Verification Results

The pre-desktop verification analysis identified 62,635.196 acres of highly suitable alienable land based on automated suitability modeling. This stage provided a broad, macro-level identification of potential areas using a 20-meter raster resolution, but lacked the spatial precision required for detailed validation.

Following systematic desktop verification, the total area retained in the inventory increased to 67,215.634 acres. This 7.3% increase is attributed to technical refinement and boundary optimization rather than any relaxation of suitability criteria.

The key factors contributing to this increase include:

- **Geometric consolidation:** The initial 20-meter raster model produced fragmented or discontinuous cells due to threshold-based classification. Desktop verification enabled the consolidation of such fragmented units and correction of internal gaps where ground conditions confirmed continuity of suitable land.
- **Improved boundary delineation using high-resolution data:** The use of 1:25,000 topographic base maps and high-resolution satellite imagery enabled more accurate interpretation of landform extents. In several cases, suitable areas were found to extend beyond the boundaries identified in the automated model.
- **Refinement of search areas:** The process transformed raster-based estimates into coherent, operational spatial units suitable for planning and implementation.

Overlay analysis of the refined inventory indicates that 14,432.485 acres (approximately 21.5%) overlap with Sustainable Forest Management (SFM) areas, including Community Forests, Forest Management Units (FMUs), and High Conservation Value–Key Biodiversity Areas. These areas were retained for case-by-case institutional review and consultation rather than exclusion at this stage.

In addition, this phase introduced a qualitative classification system (OK, Near River, Existing Structure, and Cultivated), enabling differentiation between parcels ready for immediate planning and those requiring further field or legal verification.

Overall, the transition from pre- to post-desktop verification has improved data accuracy and decision readiness, resulting in a more realistic representation of on-ground conditions for sustainable land management.

Figure 4: Model Output Vs Desktop-Verified Results



Figure 5: Pre-Desktop and Desktop-Verified by Dzongkhag

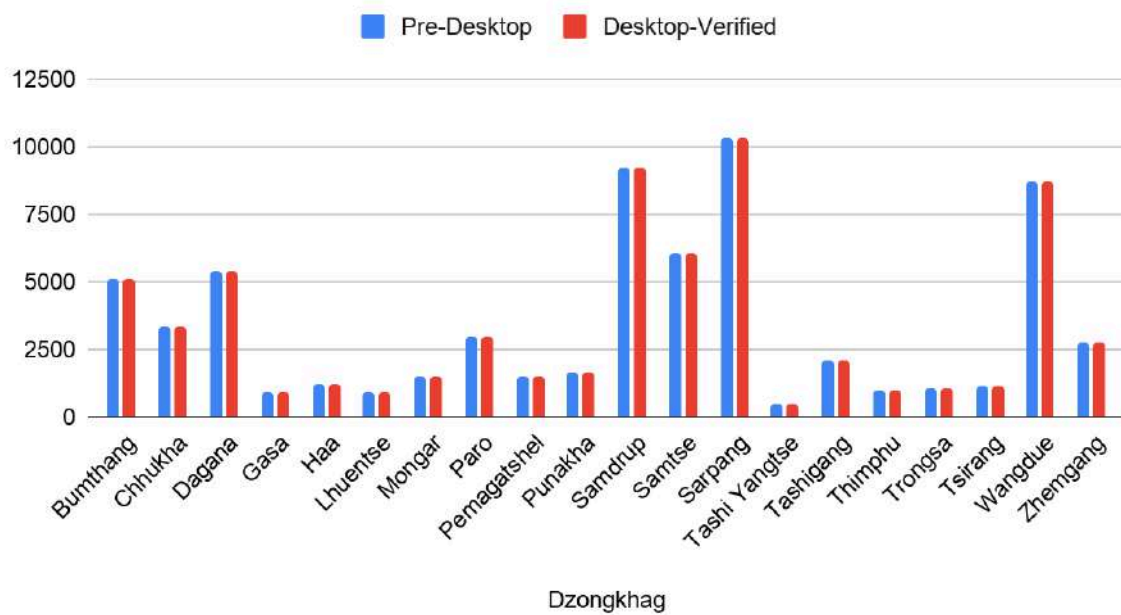


Figure 6: Desktop-Verified Plots Within SMFA versus Land Use Conflict-Free Areas (Area in Acres)

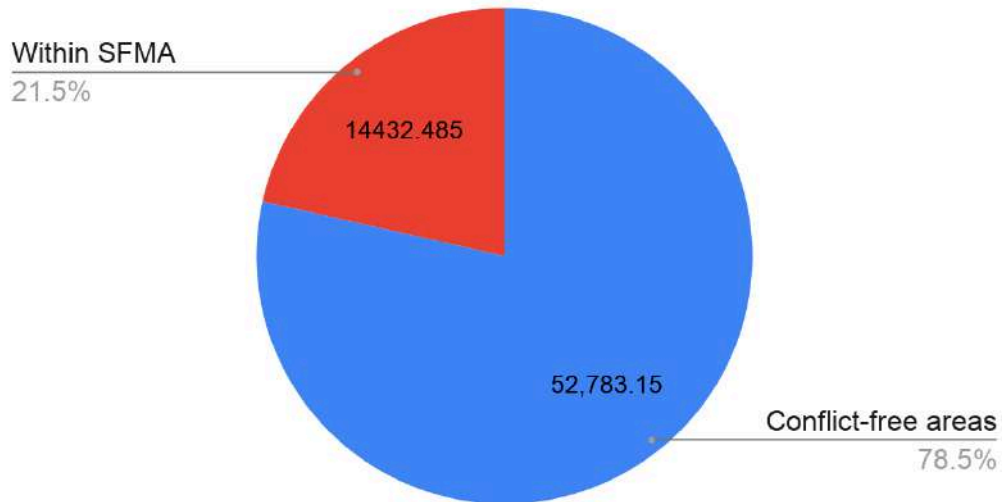
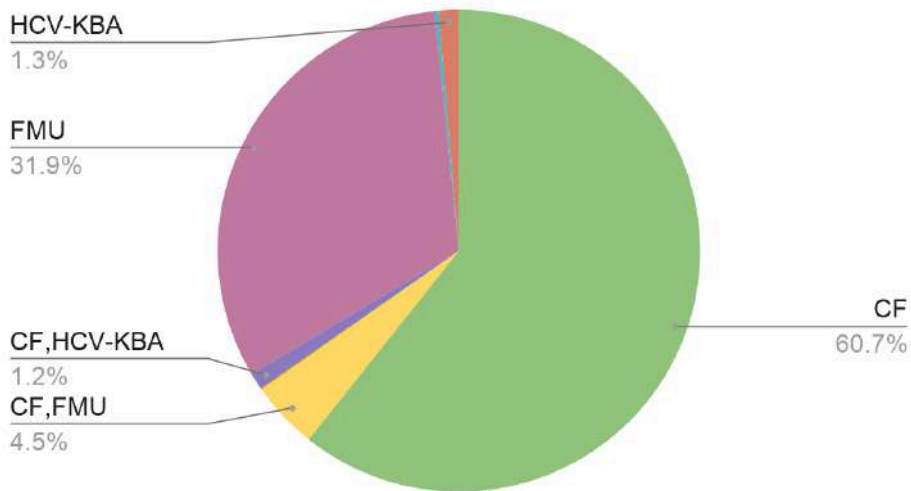


Figure 7: Distribution of Sub-Land Use in SFMA Within Desktop-Verified Areas



While desktop verification significantly improves data reliability, certain uncertainties remain that require on-ground validation. Therefore, selected parcels(sampling) were subjected to field validation to further refine and confirm their suitability.

5. VALIDATION OF DESKTOP-VERIFIED PARCELS THROUGH FIELD ASSESSMENT

5.1 Overview of Field Validation

This section presents the field validation results for selected samples from the desktop-verified, high-suitability parcels. The objective is to verify, refine, and validate the classification derived from GIS-based Multi-Criteria Decision Analysis (MCDA) and subsequent desktop verification.

Field validation was conducted across six dzongkhags, specifically Paro, Punakha, Thimphu, Wangdue Phodrang, Samtse, and Sarpang. This exercise covered a representative sample totaling approximately 3,218.966 acres. The assessment focused on confirming terrain conditions, accessibility, land use, and potential development constraints.

This stage represents the final step in transforming model-derived and visually interpreted outputs into ground-verified, decision-ready land information.

5.2 Purpose of Field Validation

The field validation was undertaken to:

- Confirm the accuracy of desktop verification classifications
- Identify on-ground constraints not visible in satellite imagery
- Refine feasibility classification of parcels
- Improve confidence in the alienable land inventory

5.3 Overall Summary

No. of Dzongkhags covered	6
No. of Gewogs covered	64
No. of sites visited	461
Total area validated (in acres)	3218.966

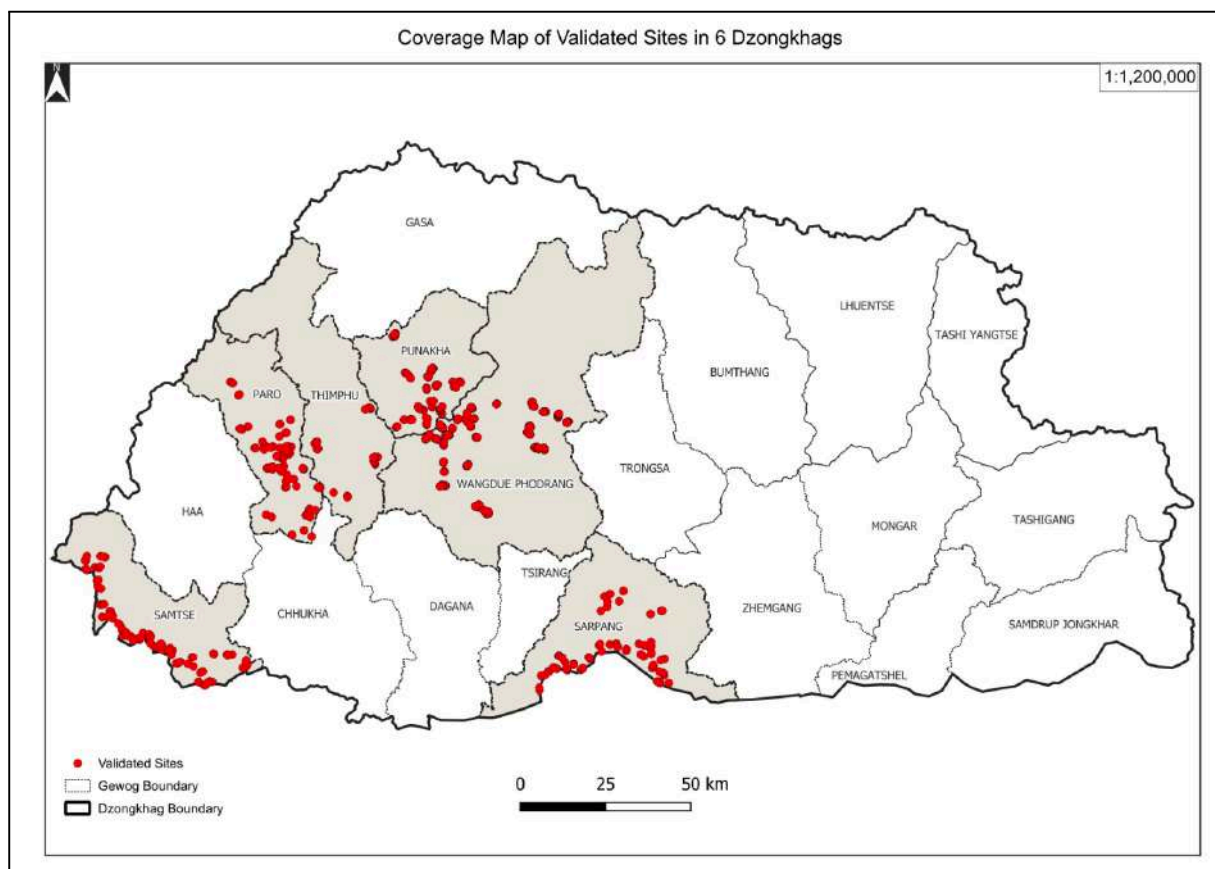


Figure 8: Overview map of field validation in 6 Dzongkhags

5.3.1 Comparative Analysis between desktop verified and field validated data

Table 4: Details before field validation

Desktop status	Count	Area(acres)
Cultivated	42	267.145
Existing structure	56	225.965
Near to river	6	45.945
ok	357	2679.911
Grand Total	461	3218.966

Table 5: Details after field validation

Field Status	Count	Area(acres)
developed/occupied /cultivated	49	216.592
Feasible	243	1650.059
Feasible but near to river	24	396.867
Not feasible	50	135.942
Others	8	164.405
Partially feasible	87	655.101
Grand Total	461	3218.966

Figure 9: Desktop status in percentage

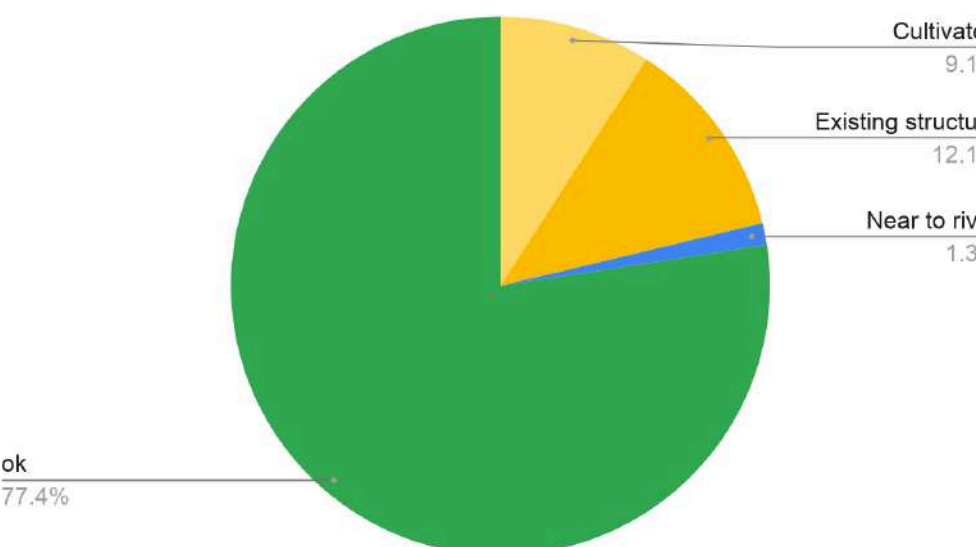
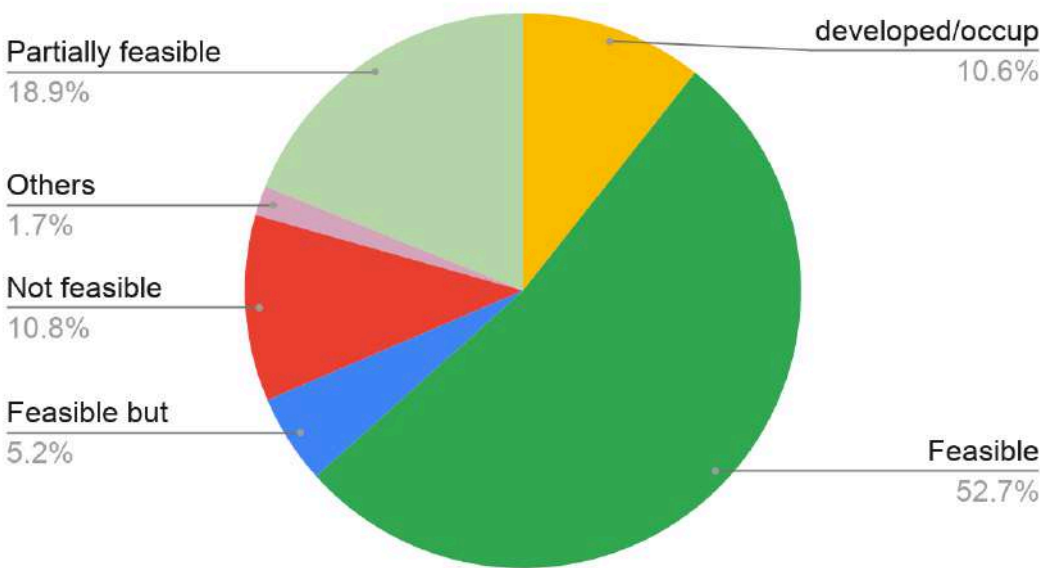


Figure 10: Status after field validation in percentage

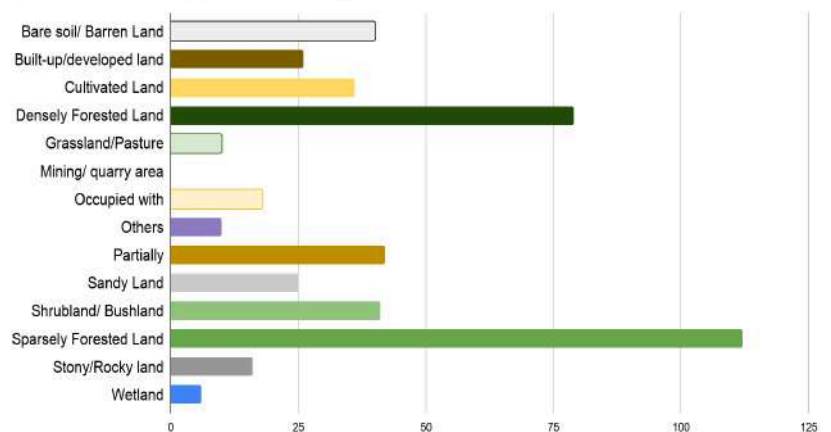


5.3.2 Ground surface status of validated sites

Table 6: Statistics of ground surface status of validated sites.

Surface status	Count	Area (acres)
Bare soil/ Barren Land	40	168.442
Built-up/developed land	26	108.463
Cultivated Land	36	157.096
Densely Forested Land	79	910.671
Grassland/Pasture	10	45.594
Mining/ quarry area	0	0
Occupied with permanent fence	18	43.418
Others	10	49.034
Partially developed/occupied	42	351.56
Sandy Land	25	297.733
Shrubland/ Bushland	41	157.747
Sparsely Forested Land	112	824.938
Stony/Rocky land	16	83.428
Wetland/Marshy	6	20.842
Grand Total	461	3218.966

Figure 11: Count of ground surface type of field validated sites



Analysis Summary: Desktop Assessment vs. Field Ground-Truthing

The analysis of 3,218.966 acres across 461 sites highlights the essential role of ground-truthing in land assessment. While the Desktop Status provided a preliminary foundation, initially identifying 77.4% of sites as “OK”, it primarily represented a measure of potential rather than certainty. Field validation refined this assessment by converting preliminary confidence into technical certainty. The results show that 52.7% of sites are fully “Feasible,” while a significant 18.9% are only “Partially Feasible,” and 5.2% are constrained due to their proximity to rivers.

This process effectively identified critical physical and social realities, including the fact that 58.8% of the area is currently under forest or shrub cover and 20.5% is encumbered by cultivation or existing structures. Ultimately, field validation transformed the inventory from a broad desktop estimate into a verified and reliable database. While this represents a generic field validation, more specific land-use suitability should be determined by applying need-specific parameters and conducting detailed topographic surveys.

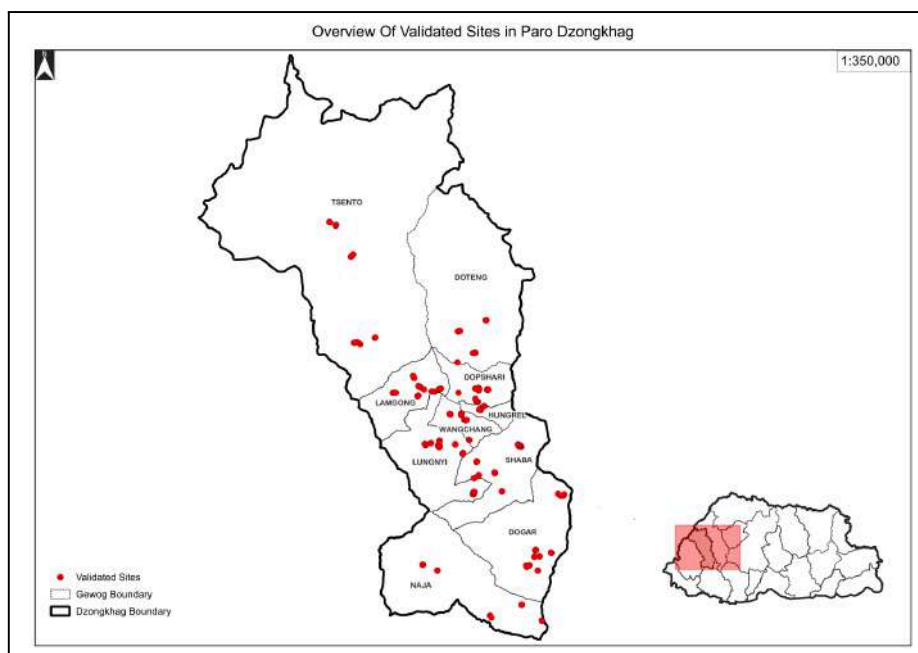
5.4 Dzongkhag-wise Field Sampling Assessment

5.4.1 Paro Field Report

Summary of Validated Plots

Total plots validated	90
Total area in acres	560.127
Number of gewogs in dzongkhag	10
Number of gewogs validated	10

Figure 12: Overview map of validated sites of Paro Dzongkhag



Geographic information about Paro Dzongkhag

Paro Dzongkhag is in northwest Bhutan, known for its fertile valley, Paro International Airport, and rich culture, with altitudes from 2,000 to over 5,600m, bordered by Thimphu, Haa, and China, featuring the Pa Chhu river, and boasting high forest cover with agriculture and tourism as key economic drivers.

Comparative Analysis of Desktop Verified vs Field Validated data

Figure 13: Statistics of Desktop verified data selected for field validation of Paro Dzongkhag

Desktop status	Count	Area(acres)
ok	84	533.908
Near to river	0	0
Cultivated	0	0
Existing structure	6	26.219

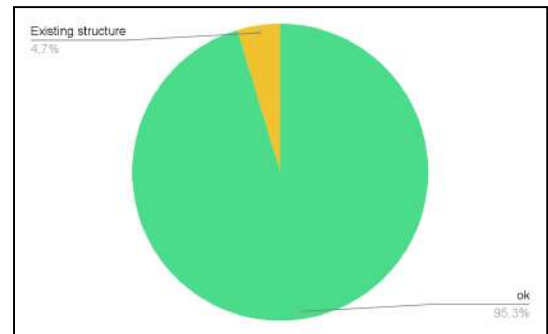


Figure 14: Provisional Land Suitability statistics after field validation of Paro Dzongkhag

Land Use Feasibility	Count	Area(acres)
Feasible	41	328.002
Feasible but near to river	1	1.756
Partially feasible	27	147.648
developed/occupied/cultivated	15	56.704
Not feasible	4	3.381
Others	2	22.636

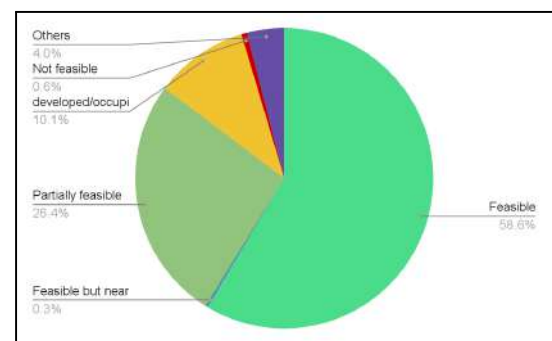


Figure 15 : Comparison between Desktop Verified vs Field Validated of highly feasible sites of Paro Dzongkhag by count

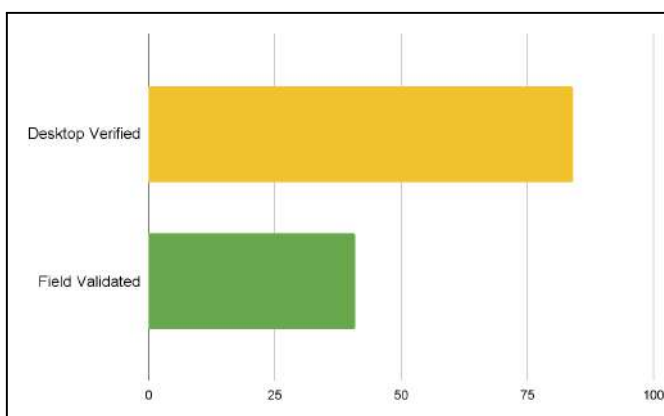
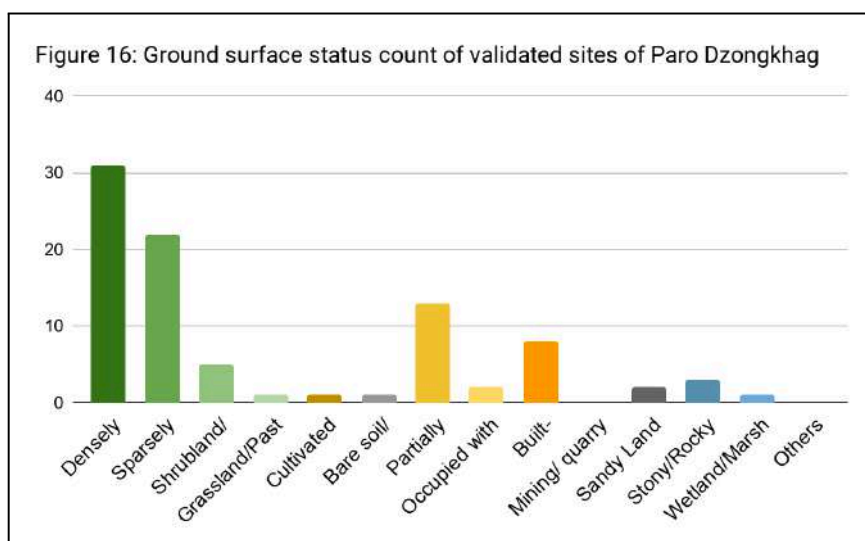


Table 7: Ground Surface status as observed during validation of Paro Dzongkhag

Surface status	Count	Area
Densely Forested Land	31	304.95
Sparsely Forested Land	22	102.384
Shrubland/ Bushland	5	12.708
Grassland/Pasture	1	3.227
Cultivated Land	1	1.009
Bare soil/ Barren Land	1	1.063
Partially developed/occupied	13	61.868
Occupied with permanent fence	2	2.611
Built-up/developed land	8	43.897
Mining/ quarry area	0	0
Sandy Land	2	2.777
Stony/Rocky land	3	22.889
Wetland/Marshy	1	0.744
Others	0	0

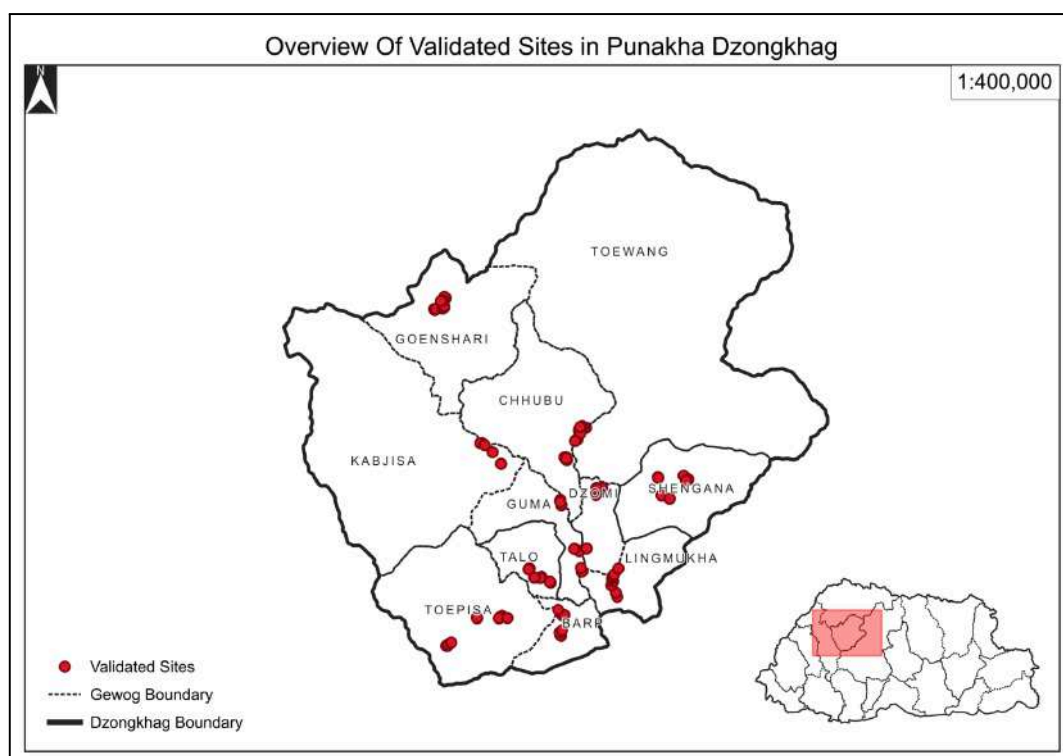


5.4.2 Punakha Field Report

Summary of Validated Plots

Total plots validated	74
Total area in acres	216.280
Number of gewogs in dzongkhag	11
Number of gewogs validated	11

Figure 17: Overview map of validated sites of Punakha Dzongkhag



Geographic information about Punakha Dzongkhag

Punakha Dzongkhag covers an area of about 1,100 km² and has a population of approximately 28,000 people. The population is largely rural, with most settlements concentrated in the fertile river valleys along the Pho Chhu and Mo Chhu, where agriculture is the main livelihood.

Comparative Analysis of Desktop Verified vs Field Validated data

Figure 18: Statistics of Desktop verified data selected for field validation of Punakha Dzongkhag

Desktop status	Count	Area(acres)
ok	57	166.578
Near to river	1	1.868
Cultivated	6	27.542
Existing structure	10	20.292

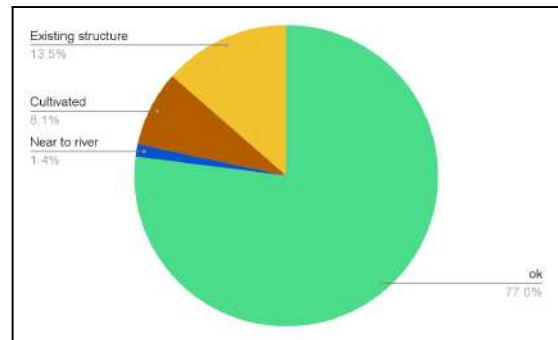


Figure 19: Provisional Land Suitability statistics after field validation of Punakha Dzongkhag

Land Use Feasibility	Count	Area(acres)
Feasible	48	167.341
Feasible but near to river	8	17.744
Partially feasible	5	10.239
developed/occupied /cultivated	3	4.112
Not feasible	10	16.844
Others	0	0

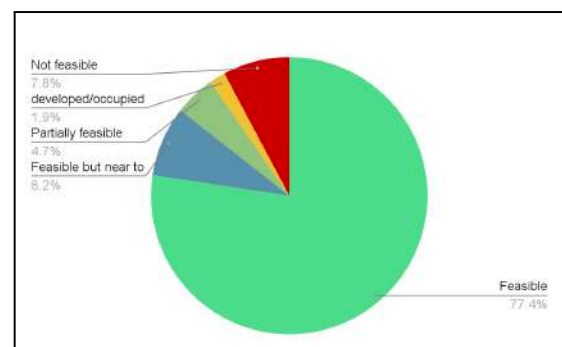


Figure 20 : Comparison between Desktop Verified vs Field Validated of highly feasible sites of Punakha Dzongkhag by count

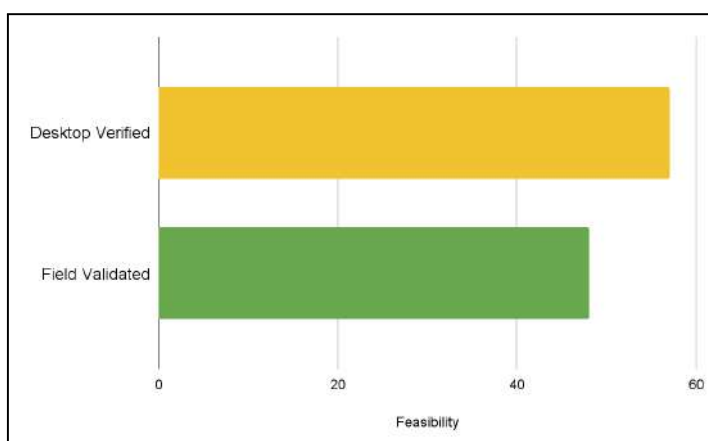
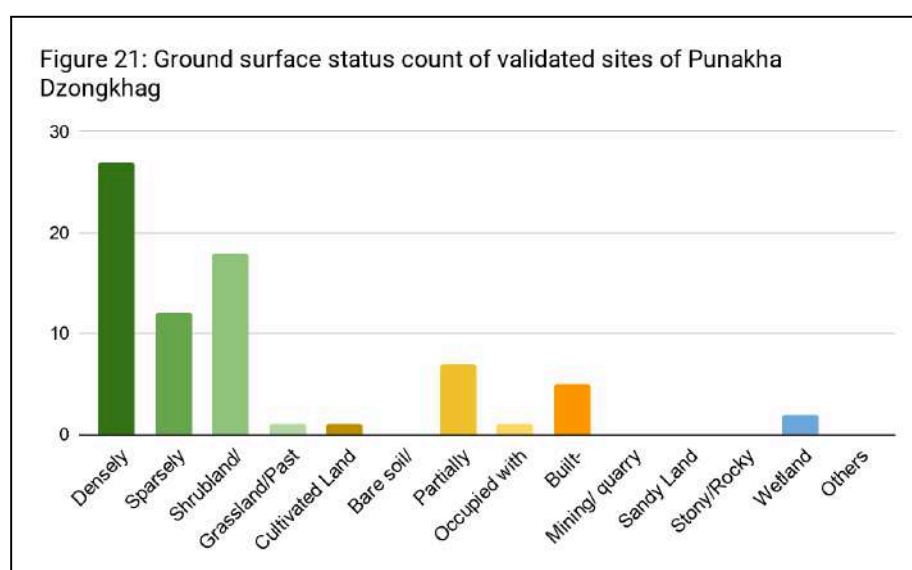


Table 8: Ground Surface status as observed during validation of Punakha Dzongkhag

Surface status	Count	Area
Densely Forested Land	27	87.697
Sparsely Forested Land	12	48.614
Shrubland/ Bushland	18	30.925
Grassland/Pasture	1	3.680
Cultivated Land	1	3.319
Bare soil/ Barren Land	0	0
Partially developed/occupied	7	10.200
Occupied with permanent fence	1	1.270
Built-up/developed land	5	12.602
Mining/ quarry area	0	0
Sandy Land	0	0
Stony/Rocky land	0	0
Wetland	2	17.973
Others	0	0

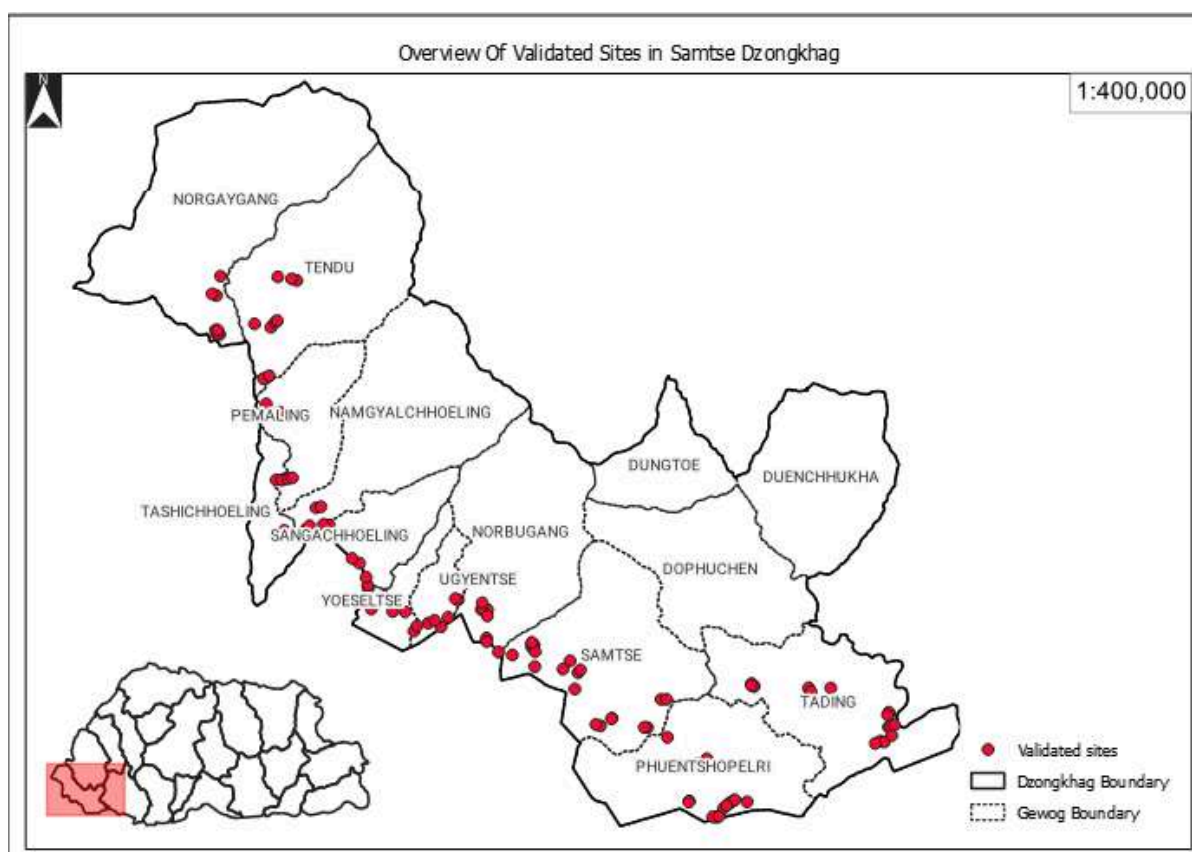


5.4.3 Samtse Field Report

Summary of Validated Plots

Total plots validated	113
Total area in acres	714.393
Number of gewogs in dzongkhag	15
Number of gewogs validated	12

Figure 22: Overview map of validated sites in Samtse Dzongkhag



Geographic information about Samtse Dzongkhag

Samtse Dzongkhag, in southwestern Bhutan bordering India, is a 1,300 km² area of hills and river valleys (200–2,000 m) with a subtropical climate, forests and agricultural land, drained by the Torsa River, and connected by southern highways.

Comparative Analysis of Desktop Verified vs Field validated data

Figure 23: Statistics of Desktop verified data selected for field validation of Samtse Dzongkhag

Desktop status	Count	Area(acres)
ok	86	595.174
Near to river	4	42.458
Cultivated	14	56.225
Existing structure	9	20.536

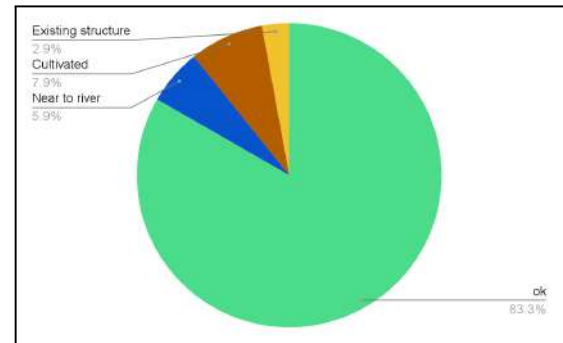


Figure 24: Provisional Land Suitability statistics after field validation of Samtse Dzongkhag

Land Use Feasibility	Count	Area(acres)
Feasible	59	332.52
Feasible but near to river	4	44.991
Partially feasible	32	251.908
developed/occupied/cultivated	8	47.199
Not feasible	8	32.879
Others	2	4.896

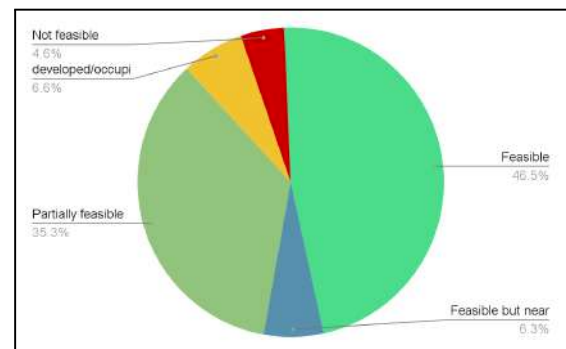


Figure 25 : Comparison between Desktop Verified vs Field Validated of highly feasible sites of Samtse Dzongkhag by count

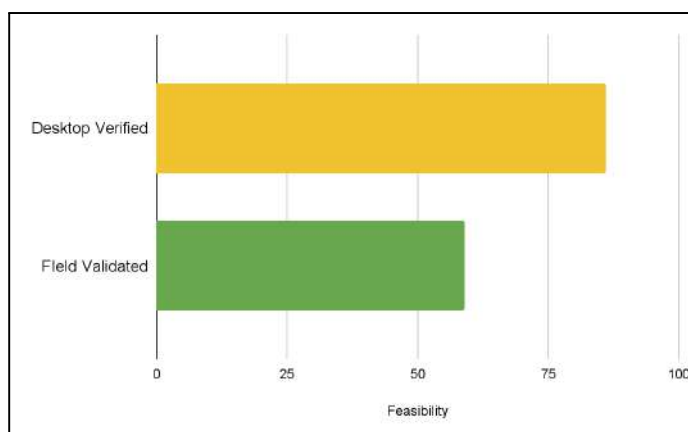
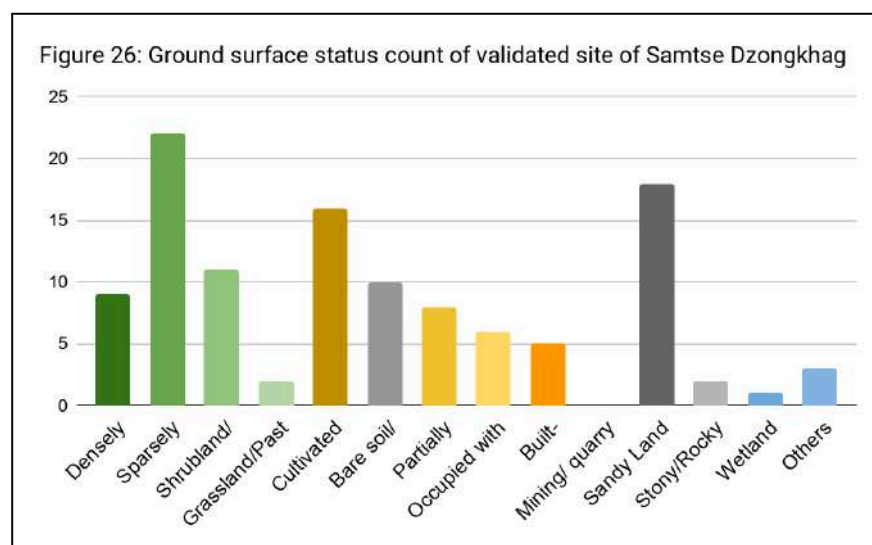


Table 9: Ground Surface status as observed during validation of Samtse Dzongkhag

Surface status	Count	Area
Densely Forested Land	9	118.524
Sparsely Forested Land	22	141.002
Shrubland/ Bushland	11	61.684
Grassland/Pasture	2	12.98
Cultivated Land	16	47.739
Bare soil/ Barren Land	10	31.857
Partially developed/occupied	8	46.629
Occupied with permanent fence	6	16.662
Built-up/developed land	5	17.724
Mining/ quarry area	0	0
Sandy Land	18	208.285
Stony/Rocky land	2	5.222
Wetland	1	0.717
Others	3	5.368

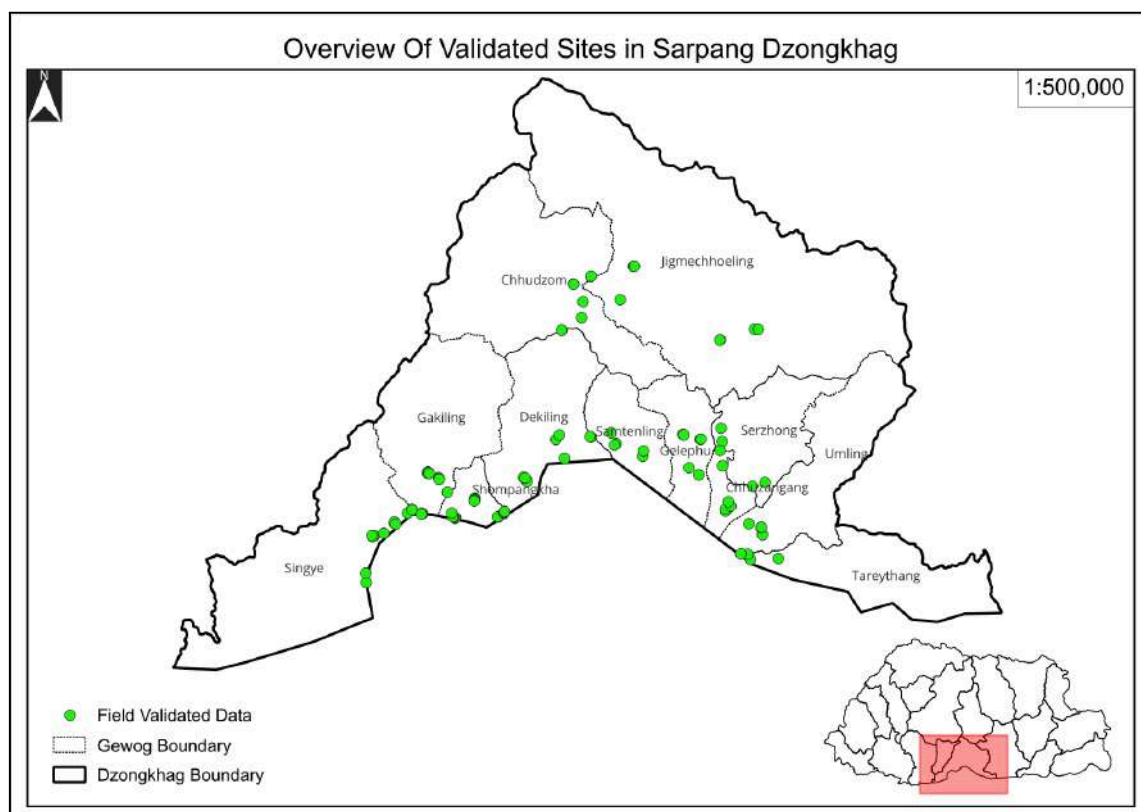


5.4.4 Sarpang Field Report

Summary of Validated Plots

Total plots validated	61
Total area in acres	1146.074
Number of gewogs in dzongkhag	12
Number of gewogs validated	12

Figure 27: Overview map of validated sites of Sarpang Dzongkhag



Geographic information about Sarpang Dzongkhag

Sarpang Dzongkhag is located in the southern part of Bhutan, along the international border with the Indian state of Assam. It is one of Bhutan's most geographically diverse dzongkhags, ranging from low-lying subtropical plains in the south to hilly and rugged terrain in the north. Sarpang plays an important role as a gateway between southern Bhutan and India, with Gelephu serving as a major urban and economic center.

Comparative Analysis of Desktop Verified vs Field Validated Data

Figure 28: Statistics of Desktop verified data selected for field validation of Sarpang Dzongkhag

Desktop status	Count	Area(acres)
Ok	46	1064.103
Near to river	0	0
Cultivated	10	69.852
Existing structure	5	12.119

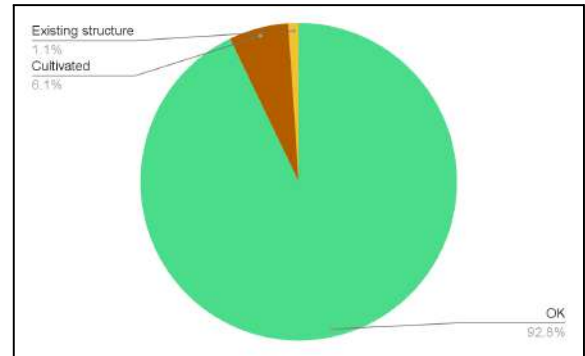


Figure 29: Provisional Land Suitability statistics after field validation of Sarpang Dzongkhag

Land Use Feasibility	Count	Area(acres)
Feasible	19	355.303
Feasible but near to river	7	321.905
Partially feasible	9	191.415
developed/occupied/cultivated	18	103.406
Not feasible	4	37.172
Others	4	136.873

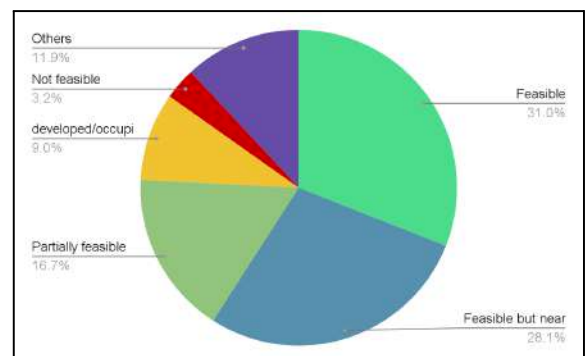


Figure 30 : Comparison between Desktop Verified vs Field Validated of highly feasible sites of Sarpang Dzongkhag by count

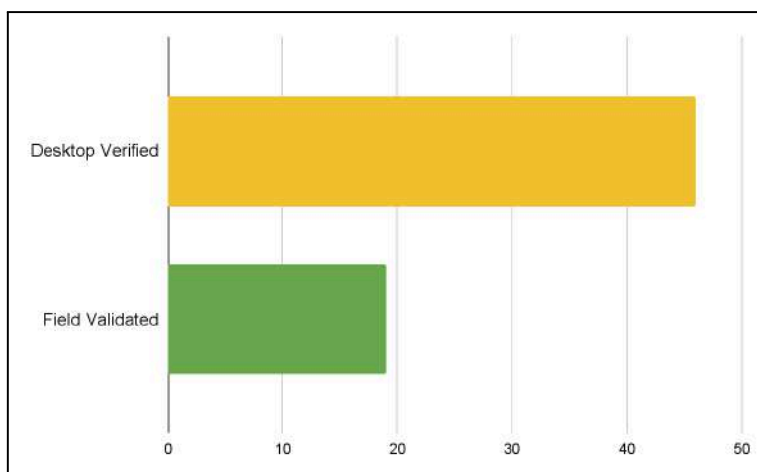
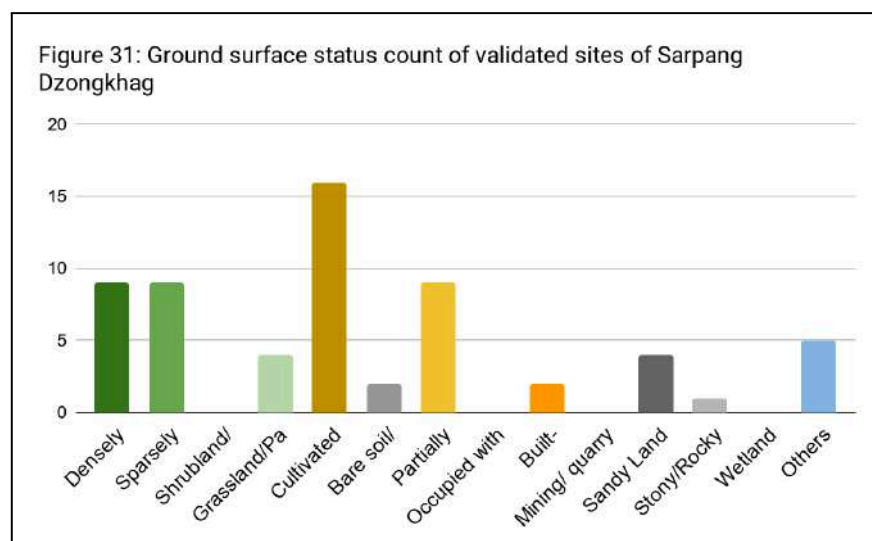


Table 10: Ground Surface status as observed during validation of Sarpang Dzongkhag

Surface status	Count	Area
Densely Forested Land	9	393.674
Sparsely Forested Land	9	238.448
Shrubland/ Bushland	0	0
Grassland/Pasture	4	16.425
Cultivated Land	16	101.141
Bare soil/ Barren Land	2	16.687
Partially developed/occupied	9	224.764
Occupied with permanent fence	0	0
Built-up/developed land	2	7.663
Mining/ quarry area	0	0
Sandy Land	4	85.362
Stony/Rocky land	1	32.263
Wetland	0	0
Others	5	29.647

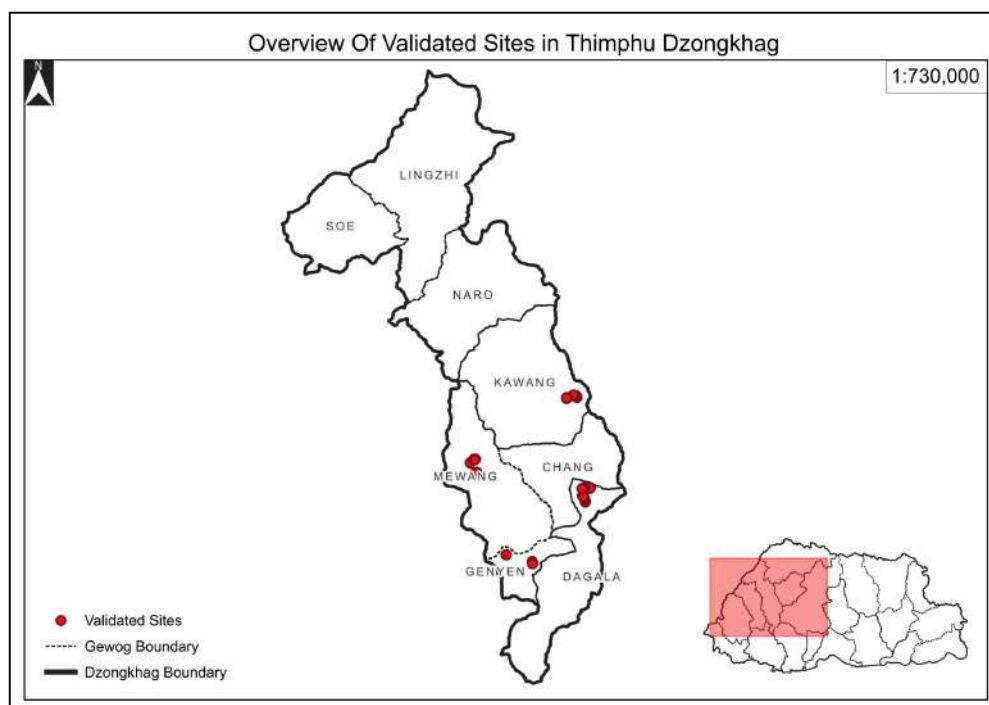


5.4.5 Thimphu Field Report

Summary of Validated Plots

Total plots validated	24
Total area in acres	59.028
Number of gewogs in dzongkhag	8
Number of gewogs validated	4

Figure 32: Overview map of validated sites of Thimphu Dzongkhag



Geographic information about Thimphu Dzongkhag

Thimphu Dzongkhag covers an area of about 1,748 km² and has a population of approximately 138,000 people. It is the most urbanized dzongkhag in Bhutan, with a large share of the population concentrated in and around Thimphu city, while smaller rural settlements are scattered across the surrounding valleys and upland areas, where agriculture and livestock rearing remain important livelihoods.

Comparative Analysis of Desktop Verified vs Field validated data

Figure 33: Statistics of Desktop verified data selected for field validation of Thimphu Dzongkhag

Desktop status	Count	Area(acres)
ok	19	52.926
Near to river	0	0
Cultivated	1	0.997
Existing structure	4	5.105

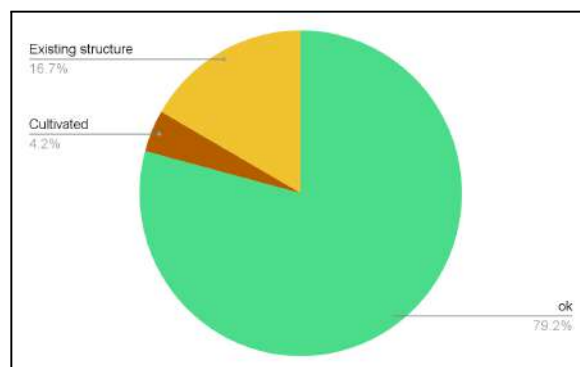


Figure 34: Provisional Land Suitability statistics after field validation of Thimphu Dzongkhag

Land Use Feasibility	Count	Area(acres)
Feasible	14	36.531
Feasible but near to river	0	0
Partially feasible	4	16.010
developed/occupied/cultivated	3	2.415
Not feasible	3	4.072
Others	0	0

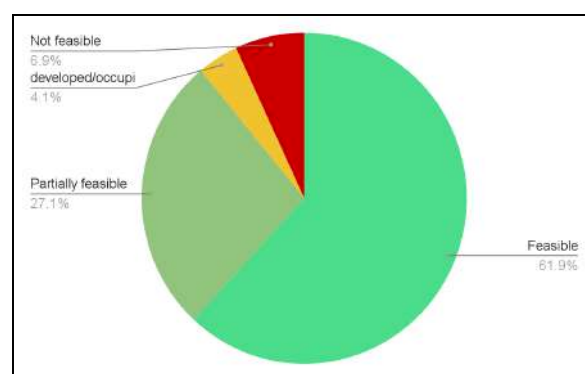


Figure 35: Comparison between Desktop Verified vs Field Validated of highly feasible sites of Thimphu Dzongkhag by count

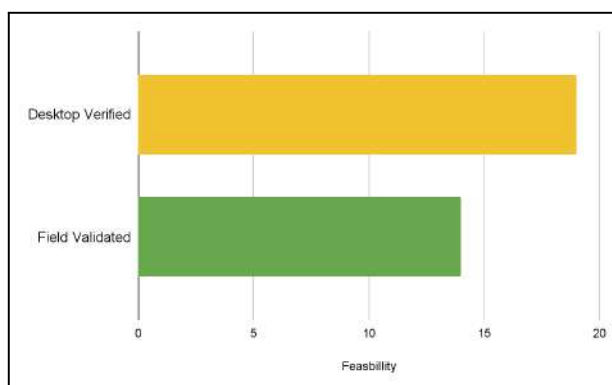
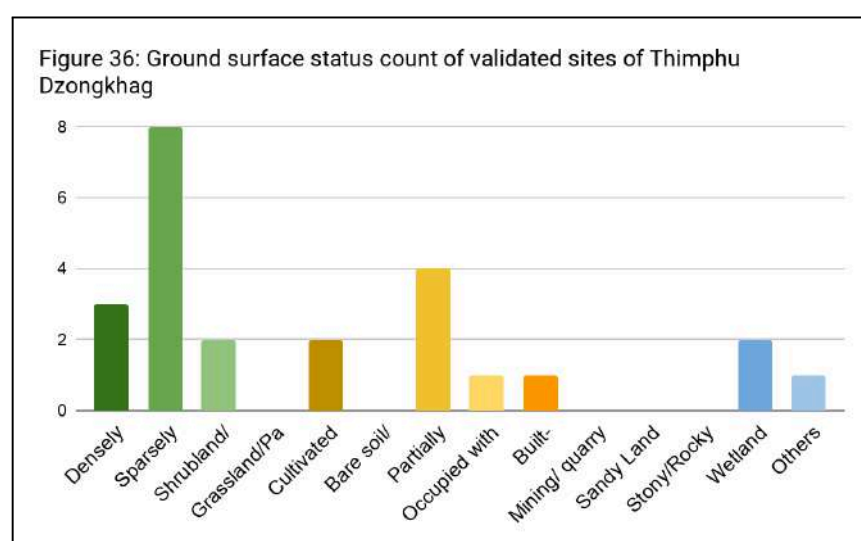


Table 11: Ground Surface status as observed during validation of Thimphu Dzongkhag

Surface status	Count	Area
Densely Forested Land	3	5.826
Sparsely Forested Land	8	34.803
Shrubland/ Bushland	2	3.214
Grassland/Pasture	0	0
Cultivated Land	2	3.888
Bare soil/ Barren Land	0	0
Partially developed/occupied	4	6.502
Occupied with permanent fence	1	0.146
Built-up/developed land	1	0.898
Mining/ quarry area	0	0
Sandy Land	0	0
Stony/Rocky land	0	0
Wetland	2	1.408
Others	1	2.343

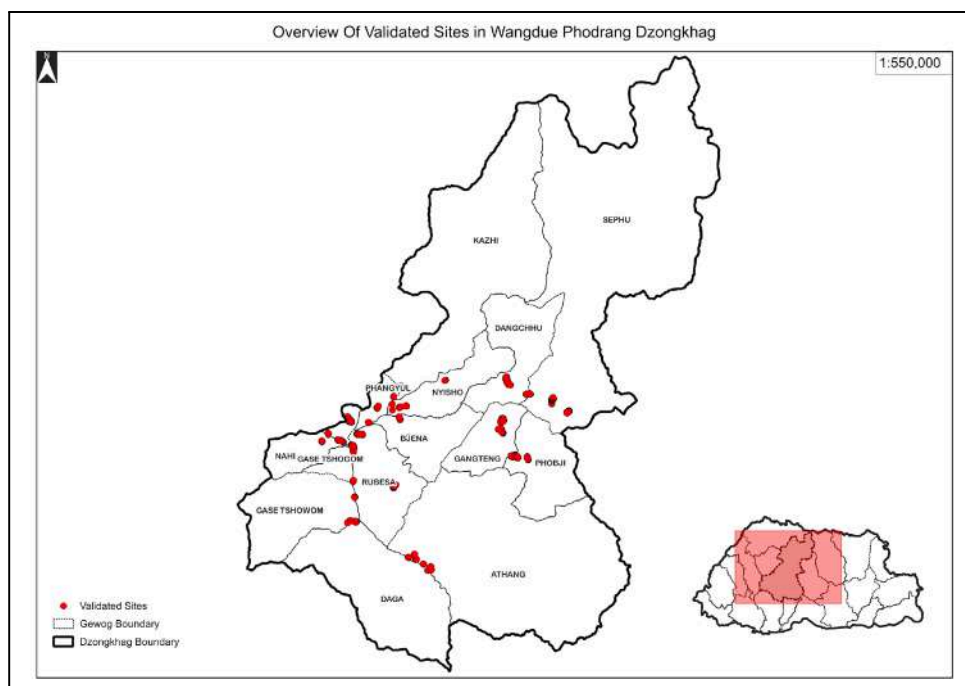


5.4.6 Wangdue Phodrang Field Report

Summary of Validated Plots

Total plots validated	99
Total area in acres	523.064
Number of gewogs in dzongkhag	15
Number of gewogs validated	15

Figure 37: Overview map of validated sites of Wangdue Phodrang Dzongkhag



Geographic information about Wangdue Phodrang Dzongkhag

Wangdue Phodrang is the largest dzongkhag in Bhutan by area (Covering over approximately 4029.03 Square Kilometers) and is bordered by Dagana and Tsirang Dzongkhags to the south, Trongsa Dzongkhag to the east, Punakha and Thimphu Dzongkhag to the west, and Gasa Dzongkhag and a small section of border with Tibet to the north. It is listed as a tentative site in Bhutan's Tentative List for UNESCO inclusion. 70.3% of the area is covered with forest.

Comparative Analysis of Desktop Verified vs Field validated data

Figure 38: Statistics of Desktop verified data selected for field validation of Wangdue Phodrang Dzongkhag

Desktop status	Count	Area(acres)
ok	65	267.222
Near to river	1	1.619
Cultivated	11	112.529
Existing structure	22	141.694

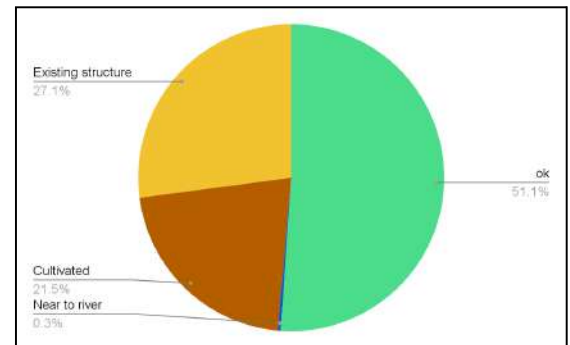


Figure 39: Provisional Land Suitability statistics after field validation of Wangdue Phodrang Dzongkhag

Land Use Feasibility	Count	Area(acres)
Feasible	62	430.362
Feasible but near to river	4	10.471
Partially feasible	10	37.881
developed/occupied /cultivated	2	2.756
Not feasible	21	41.594
Others	0	0

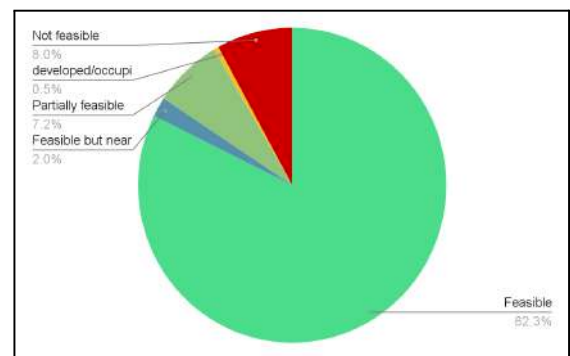


Figure 40 : Comparison between Desktop Verified vs Field Validated of highly feasible sites of Wangdue Phodrang Dzongkhag by count

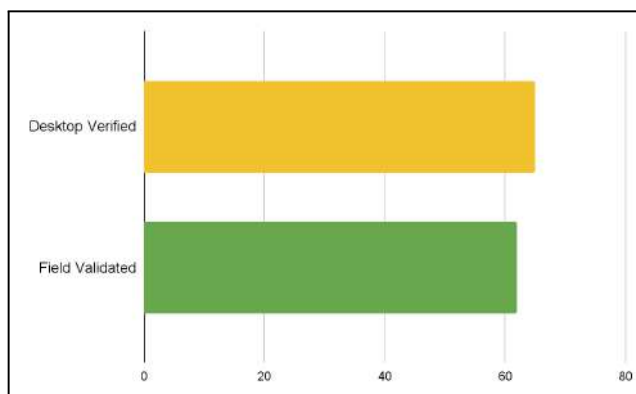
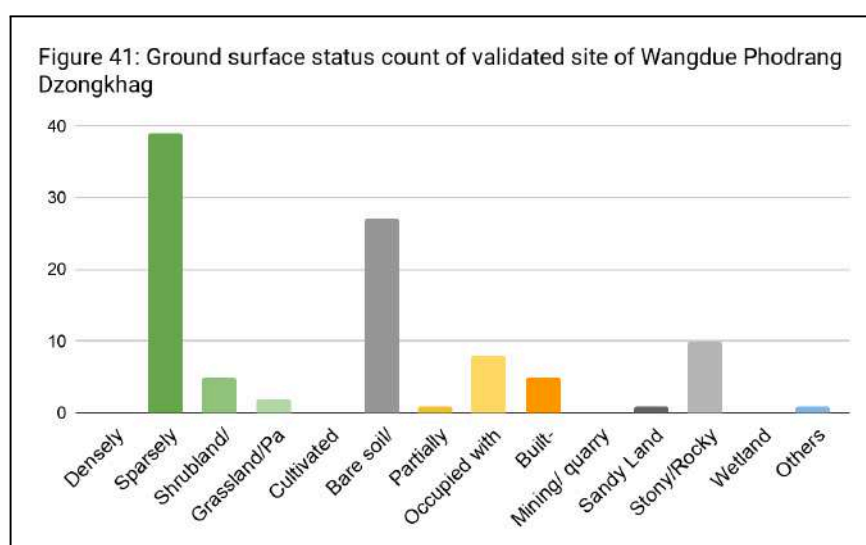


Table 12: Ground Surface status as observed during validation of Wangdue Phodrang Dzongkhag

Surface status	Count	Area
Densely Forested Land	0	0
Sparsely Forested Land	39	259.687
Shrubland/ Bushland	5	49.216
Grassland/Pasture	2	9.282
Cultivated Land	0	0
Bare soil/ Barren Land	27	118.835
Partially developed/occupied	1	1.597
Occupied with permanent fence	8	22.729
Built-up/developed land	5	25.679
Mining/ quarry area	0	0
Sandy Land	1	1.309
Stony/Rocky land	10	23.054
Wetland	0	0
Others	1	11.676



6. LIMITATIONS

While the study provides a robust validation of high-suitability land parcels, several limitations should be noted:

- Field validation was limited to selected dzongkhags and does not cover all identified parcels.
- Certain environmental risk factors such as flood and landslide susceptibility were not fully integrated due to data limitations.
- Desktop verification relies on visual interpretation of satellite imagery, which may not capture recent ground-level changes.
- Temporal differences between datasets and field conditions may affect accuracy.

These limitations should be considered when using the results for planning and decision-making.

7. CONCLUSION

This study demonstrates that GIS-based Multi-Criteria Decision Analysis (MCDA) is a valuable and effective tool for the preliminary identification of alienable land, enabling systematic and large-scale screening of potential sites. Desktop verification of GIS outputs further improves data reliability by refining preliminary classifications and identifying obvious constraints. However, the findings clearly establish that ground-truthing is indispensable for converting GIS-based potential into technical certainty, as field validation captures critical physical, social, legal, and environmental realities that cannot be fully identified through desktop analysis alone.

Based on the field validation results, it is recommended that only sites classified as *Feasible* and not densely forested be prioritized for immediate planning for any developmental activities. Sites identified as *Partially Feasible* or constrained by proximity to rivers, forest cover, or existing land use should undergo need-specific parameter analysis and detailed topographic, hydrological, and environmental surveys before any development decision is made. Institutionalizing an integrated workflow that combines GIS-based MCDA, desktop verification, and mandatory field validation will ensure sustainable and strategic land management to support the Spatial Planning Framework for Integrated Land Use and, ultimately, the National Land Use Zoning.

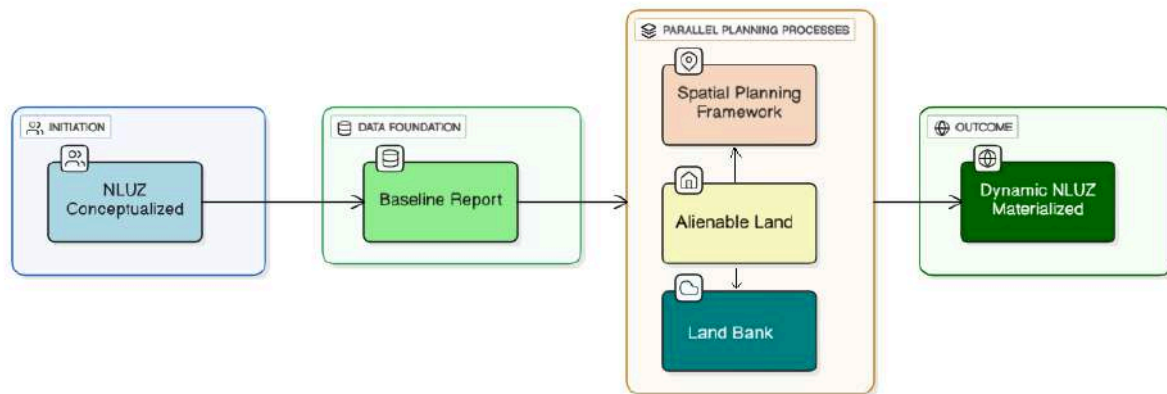


Figure 42: Conceptual cycle of NLUZ

Appendix-I:

MCDA Re-parameterization and Workflow

A Multi-Criteria Decision Analysis (MCDA) was conducted in TerrSet using the process outlined in the referenced report version. General factors such as slope, Digital Elevation Model (DEM), road proximity, Tree Crown Cover, water sources, and existing settlements were used in the analysis. The following factor weightages were applied in the MCDA conducted in TerrSet:

Slope in Degrees			
Min	Max	Value	Remarks
0	12	5	Very High Suitability
12	15	4	High Suitability
15	25	3	Moderate Suitability
25	35	2	Low Suitability
> 35		1	Very Low/Unsuitable

Water Source River			
Min	Max	Value	Remarks
0	500	5	Very High Suitability
500	1000	4	High Suitability
1000	2000	3	Moderate Suitability
2000	3000	2	Low Suitability
> 3000		1	Very Low/Unsuitable

Tree Crown Cover			
Min	Max	Value	Remarks
0	30	5	No Forest
30	39	4	Open Forest
39	59	3	Moderately Dense Forest
59	69	2	Dense Forest
69	100	1	Very Dense Forest

Elevation			
Min	Max	Value	Remarks
100	1200	5	Very High Suitability
1200	2400	4	High Suitability
2400	3000	3	Moderate Suitability
3000	3600	2	Low Suitability
> 3600		1	Very Low/Unsuitable

Road			
Min	Max	Value	Remarks
0	700	5	Very High Suitability
700	1000	4	High Suitability
1000	2000	3	Moderate Suitability
2000	3000	2	Low Suitability
> 3000		1	Very Low/Unsuitable

Settlement Proximity			
Min	Max	Value	Remarks
0	700	5	Very High Suitability
700	1000	4	High Suitability
1000	2000	3	Moderate Suitability
2000	3000	2	Low Suitability
> 3000		1	Very Low/Unsuitable

After the output was generated from TerrSet as a raster, it was vectorized in QGIS. The data was then dissolved, and only Class 5, representing highly suitable areas, was selected and exported. The constraint from Option 3 was subsequently erased from the output. Finally, automated cleaning was carried out.

Generalization and Removal of Area less than 1 acre

The smoothing of the sharp edges of the polygons produced was carried out in QGIS so that they could be generalized. To achieve this, a buffer of 10 m was applied to the resultant data, followed by a dissolve operation, and then a negative buffer of 10 m was applied. The constraint from Option 3 was then erased from the resultant data. After that, a multipart-to-singlepart process was performed to create individual shapefiles instead of grouped complex ones. Fix Geometries was applied to the resultant data. Finally, using the Field Calculator, the areas were recalculated in QGIS, and polygons with areas less than 1 acre were removed.

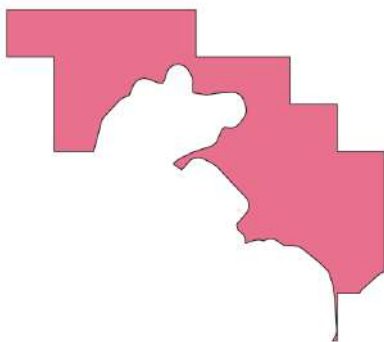


Figure Anx-I-1: Before Generalization

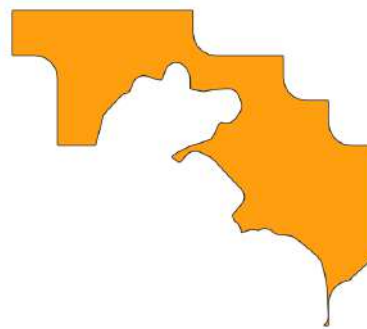


Figure Anx-I-2: After Generalization

Powerline buffer Removal

Since power lines were also important to be removed from the merged and generalized Class 5 data, a power line buffer removal was applied to the dataset. The intersection of the power line buffer with the generalized Class 5 data was selected using the Select by Location tool in QGIS. The power line buffer areas were then removed from the intersected data using the Difference tool. After that, automated data cleaning was applied. Finally, the cleaned data was merged with the generalized Class 5 dataset.



Figure Anx-I-3: Before Powerline Removal

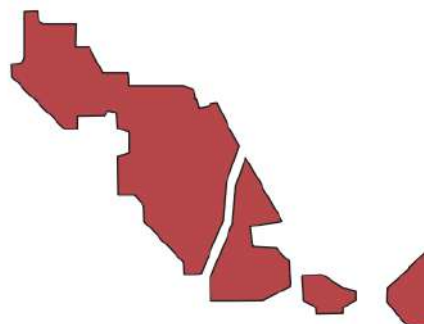


Figure Anx-I-4: After Powerline Removal

Discounting Highway buffer(Right of Way)

It was necessary to remove areas near the highway using the Highway RoW buffer, as this was not included in the general constraints for option 3. To achieve this, we first selected the intersections of the Highway RoW shapefile with the CL5 plots using the Select by Location tool in QGIS. Then, the Difference tool in QGIS was used to remove the areas of the shapefile overlapping with the Highway RoW. Next, the Multipart to Singleparts tool was applied to separate the polygons into individual features. The Thinness Ratio was then calculated. Finally, the resultant data was ready for desktop validation.



Figure Anx-I-5: Before removing RoW



Figure Anx-I-6: After removing RoW

Land Use Overlap

To enhance the precision of land alienation assessments, forest constraints were integrated into the analysis using Department of Forest data, including Community Forests (CF), High Conservation Value-Key Biodiversity Areas (HCV-KBA), and Forest Management Units (FMU). Using a suite of geoprocessing tools—including Clip, Multipart to Singlepart, Point on Surface, Select by Location, and Spatial Join—the overlap between alienable land and these protected zones was identified and consolidated within the attribute table. This spatial workflow, finalized through CSV conversion for data management, further enriched the dataset by calculating specific area attributes to quantify the exact extent of overlap for each forest category.

	fid	dzongkhag	plot_id	area_ac	dv_status	dv_remarks	lu_overlap ▼	overlap_ar	remarks
1	4989	Paro	A-PR-633	16.3599999999...	ok	NULL	HCV-KBA	16.3599999999...	NULL
2	10326	Tsirang	A-TS-190	11.3729999999...	ok	NULL	HCV-KBA	11.3729999999...	NULL
3	3075	Dagana	A-DG-456	14.4890000000...	ok	NULL	HCV-KBA	9.91499999999999	NULL
4	5041	Paro	A-PR-685	53700000000000	ok	NULL	HCV-KBA	53700000000000	NULL
5	12383	Zhemgang	A-ZG-323	17600000000000	ok	NULL	HCV-KBA	5.59500000000000	NULL
6	10363	Tsirang	A-TS-227	34100000000000	ok	NULL	HCV-KBA	5.34100000000000	NULL
7	4419	Paro	A-PR-63	36700000000000	ok	NULL	HCV-KBA	5.26600000000000	NULL
8	7154	Samtse	A-ST-705	30100000000000	ok	no check	HCV-KBA	5.13700000000000	NULL
9	5285	Paro	A-PR-929	90200000000000	ok	NULL	HCV-KBA	4.90200000000000	NULL
10	10361	Tsirang	A-TS-225	82400000000000	ok	NULL	HCV-KBA	4.82400000000000	NULL
11	4988	Paro	A-PR-632	4.	ok	NULL	HCV-KBA	4.	NULL

Figure Anx-I-7: Attribute table containing FMU,CF & KBA-HCV overlap and area as lu_overlap and overlap_ar

Appendix-II:

Standard Operating Procedure (SOP) for the Field Validation of Desktop Identified Alienable Land

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Field SOP

1. Per gewog, a minimum of five sites needs to be validated.
2. Out of the five sites, a minimum of three sites should have an Ok status during desktop validation. The selection of these three sites should be based on larger plot size, accessibility, and inaccessibility (for example, some plots close to the road and at least one plot located a little farther from the road head).
3. A minimum of one plot should have the status of an existing structure or cultivated land (if possible, plot history related to such occupancy should be collected in the field).
4. A minimum of one plot should have the status of being near a river. For such plots, the elevation difference between the river and the plot needs to be recorded. These plots should also be verified, and recommendations should be made on whether they can be utilized or excluded from the GeoDB.
5. For all plots selected for verification, the slope needs to be recorded using the provided clinometer (generally, plots with slopes less than or equal to 30 degrees and greater than or equal to 1 degree are considered suitable). If there is noticeable undulation in the site's topography, multiple slope readings should be taken from different points and averaged for record.
6. All plots must be surveyed using any geo-mobile app or handheld GPS and recorded as per the prescribed template. If a full plot survey is not feasible, points should be collected from accessible locations to validate the plot as suitable or unsuitable. The revised area must be recorded, and a map must be prepared. The name of the geo-mobile app, as well as the brand and model of the mobile device, should be recorded. Additionally, a form will be provided and must be completed for each validation to capture more detailed attributes.
7. As part of the field validation, photographs must be taken from appropriate angles for every site. The photographs should be geotagged; if they are not geotagged, the image name should follow the format of the plot ID, an underscore, and a running number (for example, A-PR-105_1.img).
8. Upon completion of the field validation, the data must be submitted, and the field validation status should be integrated into the GeoDB. All other files should be submitted neatly using proper file and folder nomenclature.

Field Survey Checklist

- Point, line and polygon features with options to add plot id, type , status, remarks, photos or videos , distance and area.

Field validation form checklist

- Plot Id
- Slope in degrees
- Status
- Coordinates
- Observations
- Remarks

Field photos

- Photos with name as per point number 7 of SOP

Instruments used

Clinometer

A clinometer is a mechanical instrument for measuring angles of slope or elevation. The user looks through the sighting scope and aligns it with the target; the internal scale indicates the angle directly, making it compact and easy to use in the field for measuring heights, slopes, or inclines in forestry, surveying, and engineering.



Digital Clinometer

A digital clinometer is an electronic instrument used to measure angles of slope, elevation, or tilt with high precision. It typically features a digital display that shows the angle directly, often in degrees or percent grade. Clinometers are used to measure the angle of the riverbed slope or stream gradient. By sighting along the water surface or riverbank from a fixed point, the device indicates the inclination of the bed relative to horizontal.

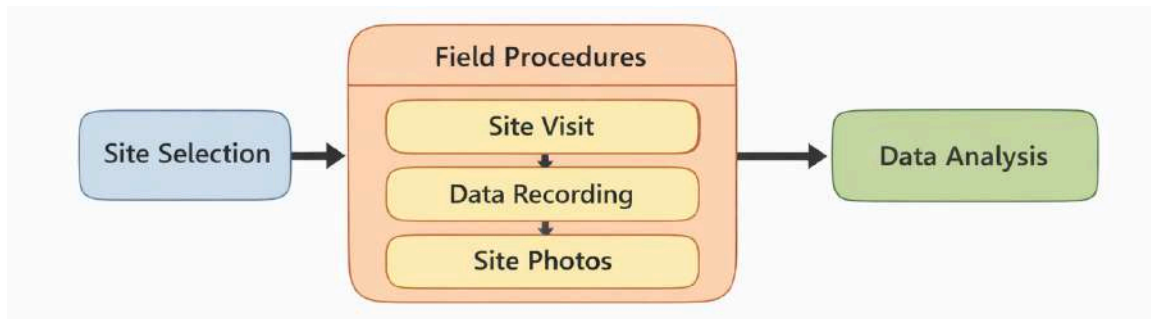


DJI Mini 4 Pro

The DJI Mini 4 Pro is a compact, lightweight drone with a 48 MP camera and 4K video, a 3-axis gimbal, and obstacle avoidance. It's used for aerial photography, videography, recreational flying, and quick inspections, offering stable, high-quality footage in a portable design.



Methodology Used During field validation in line with SOP



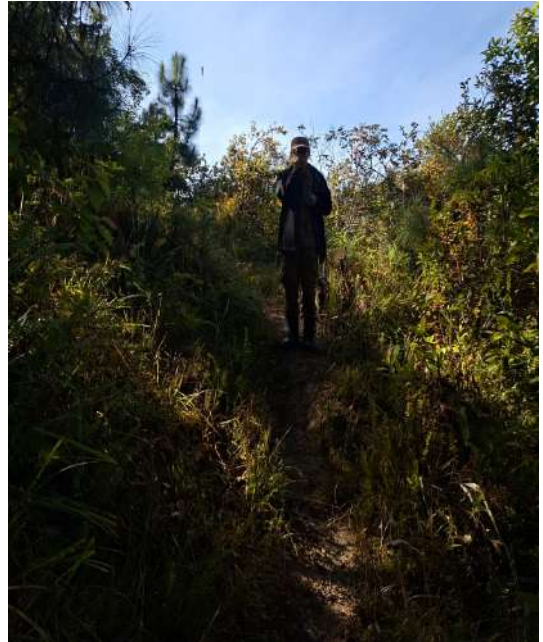
Site Selection

A minimum of 5 plots have been selected from each Gewog, with variety in plot status with preference given to larger plots and with greater coverage of the Gewog.

Field Procedures

The field team has traversed to their respective selected sites for on-field observations and data recording. The main criteria to check the site suitability were based on slope, surface status and distance from river. Clinometers were utilized to check the slope of the selected sites to ensure it fit the feasibility criteria.





All site observations and attribute data were duly recorded in the epicollect5 mobile application. Photos of the site have been recorded for reference and for data storage. The DJI Mini Pro 4 drone has also been utilized to obtain images where the team could not physically access due to terrain difficulty or to acquire images covering the entire area of interest.



Data analysis and consolidation

The spatial data, photographs, attributes, and field observations collected by each team were compiled in standardized formats, including shapefiles, KMZ, and JPG. These datasets will be consolidated and integrated into the GeoDB. A detailed field report and data analysis were also prepared. The data collected during the fieldwork are maintained in the following folder structure for archival purposes:

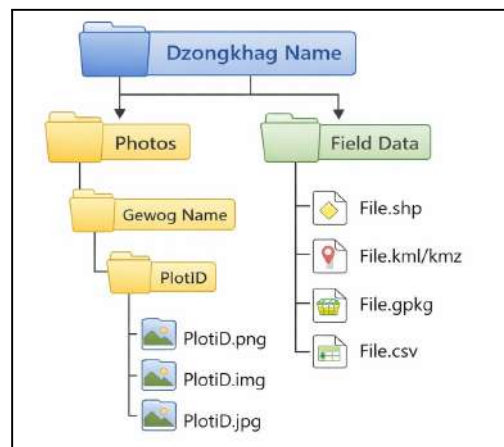


Figure 2: Folder structure for data archival

Appendix-III:

Selected Field Observation Case Studies

This appendix serves as a technical supplement to the national field validation exercise, presenting a representative selection of signature case studies (3–4 per Dzongkhag). Although 461 sites were physically validated across six Dzongkhags, only selected cases are included to demonstrate the range of ground conditions encountered and to highlight the importance of field verification in resolving data gaps, validating spatial analysis outputs, and confirming site suitability and feasibility.

1. PARO DZONGKHAG

Case 1

Plot ID	A-PR-680
Area	1.358 Acre
Location	Lamgong, Paro
Desktop Verified status	ok
Field Validated status	developed/occupied/cultivated
Slope Range in degrees	0-2
Distance from road	10m
Altitude	2330m

Plot details and Field Observations

The area is developed or occupied and it is near the river which may pose risk if settled nearby.

Location Reference



Field Photos



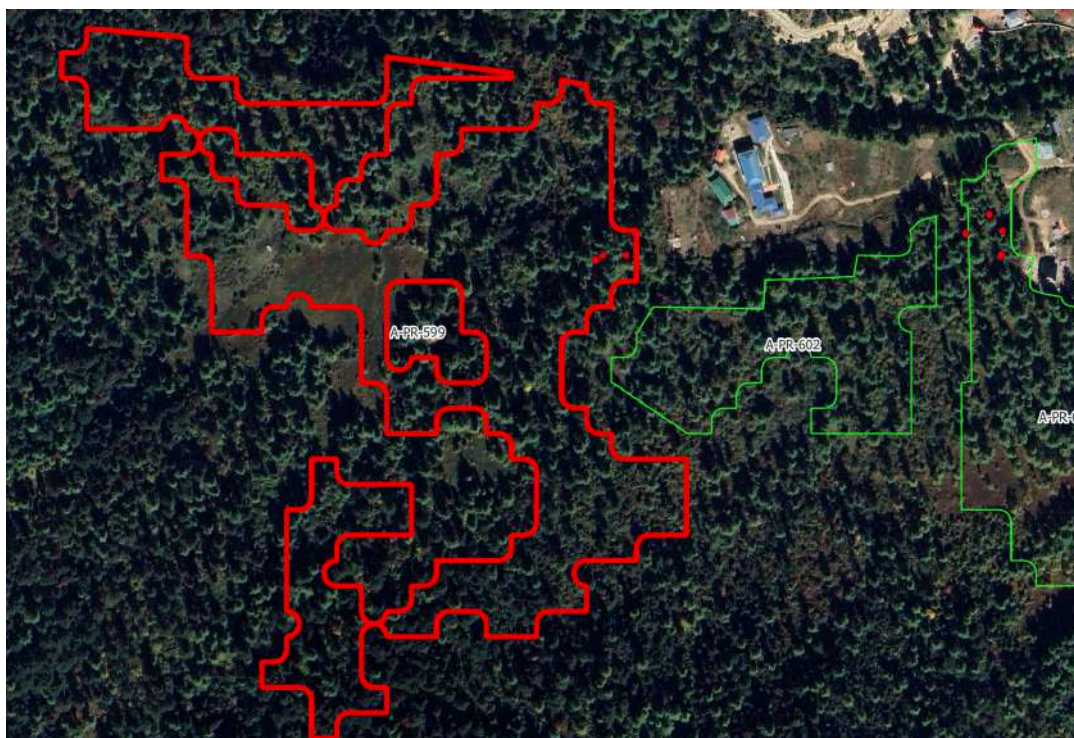
Case 2

Plot ID	A-PR-599
Area	23.09 Acre
Location	Lamgong, Paro
Desktop Verified status	ok
Field Validated status	Feasible
Slope Range in degrees	1-10
Distance from road	100m
Altitude	2594m

Plot details and Field Observations

The topography of the area seems suitable although currently it's mostly forested. As per locals, there used to be settlements in the past. Some retaining walls and ruins are still visible.

Location Reference



Field Photos



Case 3

Plot ID	A-PR-936
Area	0.16 Acre
Location	Shana, Tsento, Paro
Desktop Verified status	ok
Field Validated status	Occupied with Structure
Slope Range in degrees	1-2
Distance from road	2m
Altitude	2855m

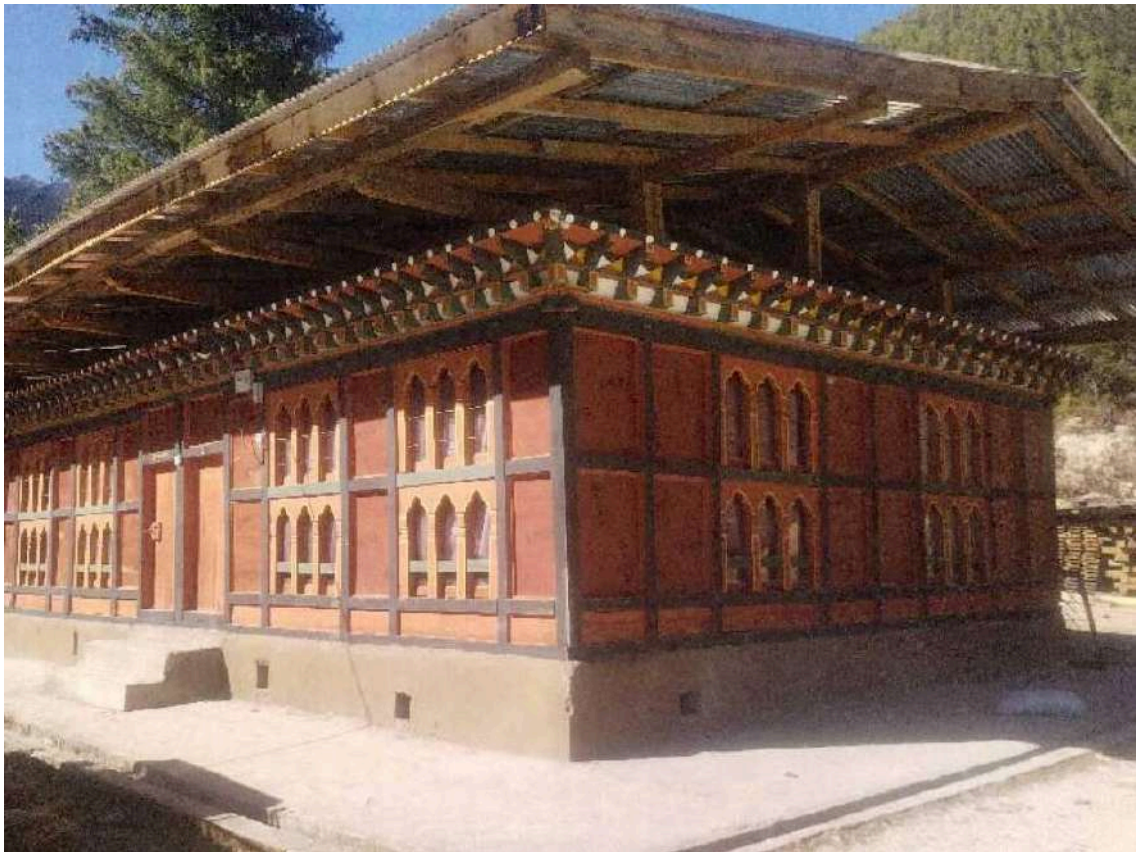
Plot details and Field Observations

There is a semi- permanent structure which is used as guest house

Location Reference



Field Photos



Case 4

Plot ID	A-PR-393
Area	5.235 Acre
Location	Dogar, Paro
Desktop Verified status	ok
Field Validated status	Feasible
Slope Range in degrees	2-16
Distance from road	300m
Altitude	2780m

Plot Details and Field Observations

The area has gentle to moderate slope, mostly covered by trees and bushes. Water is available. It's inside a permanent fence.

Location Reference



Field Photo



2. PUNAKHA DZONGKHAG

Case 1

Plot ID	A-PK-127
Area	2.604 acres
Location	Goenshari, Punakha
Desktop Verified status	OK
Field Validated status	Feasible
Slope Range in degrees	5-10
Distance from road	10m
Altitude	2299m

Plot Details and Field Observations

The area is sparsely forested with surrounding bushes. The plot is feasible and well suited for development, as it is located in a very good location close to a feeder road.

Location Reference



Field Photos



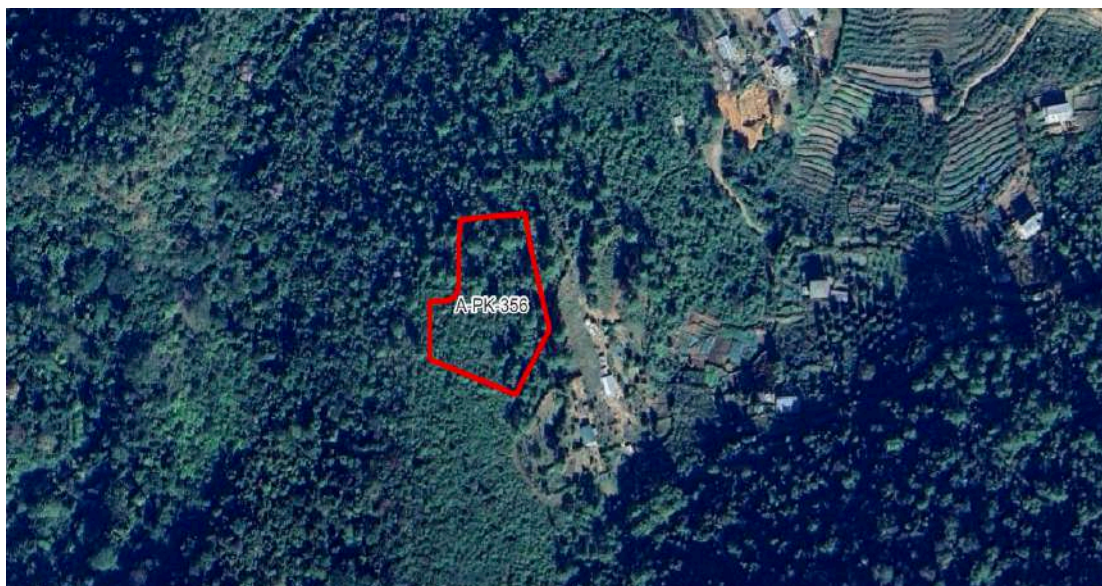
Case 2

Plot ID	A-PK-356
Area	1.576
Location	Barpisa, Punakha
Desktop Verified status	ok
Field Validated status	Feasible
Slope Range in degrees	12
Distance from road	700m
Altitude	1708m

Plot Details and Field Observations

The plot is predominantly characterized by shrubland and bushland vegetation. A few pine trees and scattered bushes are present within and around the area. Based on the field observation, the plot appears stable and accessible, making it feasible for alternative land uses such as agriculture, plantation, or other permissible developmental purposes

Location Reference



Field Photos



Case 3

Plot ID	A-PK-310
Area	1.619
Location	Talo, Punakha
Desktop Verified status	ok
Field Validated status	Densely Forested land
Slope Range in degrees	40-55
Distance from road	30m
Altitude	2043m

Plot Details and Field Observations

The plot is characterized by a steep slope and narrow terrain, which significantly limits its usability. Due to the slope gradient and restricted width, the land is not suitable or feasible for any alternative developmental, agricultural, or construction purposes.

Location Reference



Field Photos



Case 4

Plot ID	A-PK-391
Area	3.994
Location	Toedwang, Punakha
Desktop Verified status	ok
Field Validated status	feasible but near to river
Slope Range in degrees	3-5
Distance from road	500m
Altitude	1301m

Plot Details and Field Observations

The area is sparsely forested with bushes. During field validation, the plot was observed to be temporarily used by private agencies as a tourist camping site, with several camps set up on the land. The plot is flat and level, located near the riverside, making it well suited for recreational activities, ecological purposes, and picnic spots.

Location Reference



Field Photos



Case 5

Plot ID	A-PK-465
Area	1.81
Location	Lingmukha, Punakha
Desktop Verified status	ok
Field Validated status	Feasible
Slope Range in degrees	5-10
Distance from road	30m
Altitude	1790m

Plot Details and Field Observations

The plot is predominantly flat, with about 90% of the area level and covered mainly with pine trees. Surrounding areas have seen timber harvesting by local people, and the site has been reported to have water supply issues. The area has sparse vegetation and overall appears feasible for development.

Location Reference



Field Photos



3. SAMTSE DZONGKHAG

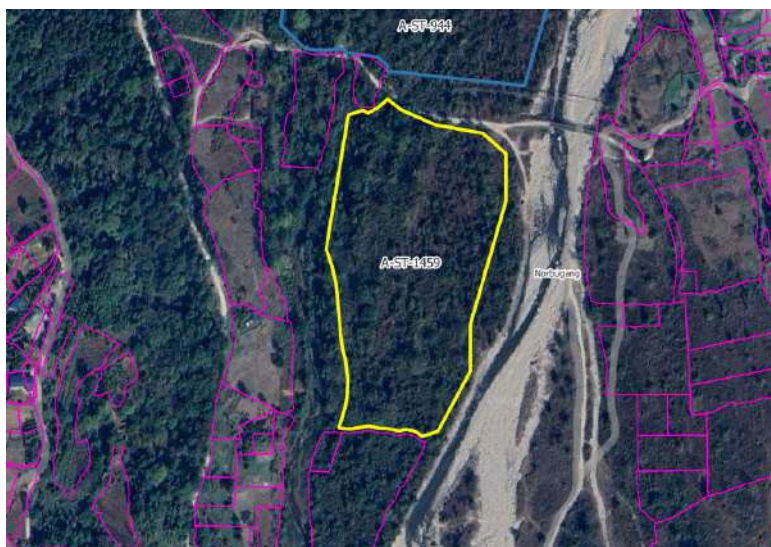
Case 1

Plot ID	A-ST-1459
Area	13.869
Location	Norbugang
Desktop Verified status	Feasible, near to river
Field Validated status	Partially develop/Occupied
Slope Range in degrees	2
Distance from road	0m
Altitude	311m

Plot details and Field Observations

The land towards the riverside is sandy in nature and is currently under development for a coal factory by SDK Construction. One of the focal persons stated that part of the alienable land has been leased to them, the project has been running for four months, and it is expected to be completed within two years. Road connectivity is new, and part of the lower section of the plots is being used as a soil and rock dumping yard. From the top right side, the current monsoon flood has inundated part of the alienable plot as well as the suspension bridge connecting Kydisa village. Although the plot is located close to the riverside and part of the adjacent plot (A-ST-944) is being damaged, development activities are observed to be in full swing.

Location Reference



Field Photos



Case 2

Plot ID	A-ST-380
Area	6.104
Location	Sampheling
Desktop Verified status	ok
Field Validated status	Partially develop/Occupied
Slope Range in degrees	5-20
Distance from road	300m
Altitude	1126m

Plot details and Field Observations

The plot is located at an altitude of 1,126 m and is approximately 400 m away from the international border at Sampheling. It is partially under active cultivation, and an outpost structure has been constructed and is currently occupied by the Royal Bhutan Army, Police, and Desuung. The remaining area consists of sparsely dense forest with bushes, and old terraces are observed on the ground, Water and electricity are available at the site

Location Reference



Field Photos



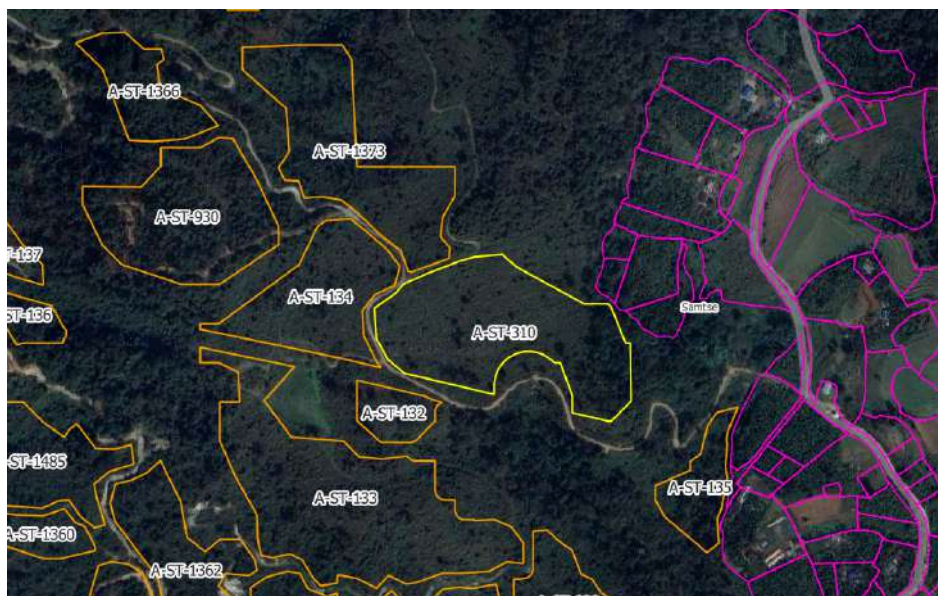
Case 3

Plot ID	A-ST-310
Area	6.455
Location	Lamaithang Toed
Desktop Verified status	ok
Field Validated status	ShrubLand/BrushLand
Slope Range in degrees	2-5
Distance from road	0m
Altitude	426m

Plot details and Field Observations

The plot is currently not under cultivation and has been left barren, covered with bushes and a few grown trees. A new farm road is connected, however, drinking water and electricity is not available. Old fallen structures or ruins are also observed on the ground in A-ST-132. The plot is found to be feasibly situated at a higher altitude compared to the village, and the slope is very favorable for cultivation and construction; however, the plot is presently covered with brush. The team also came to know that the plots are being used for cultivation.

Location Reference



Field Photos



4. SARPANG DZONGKHAG

Case 1

Plot ID	A-SP-134
Area	65.117
Location	26.924895N, 90.367633E
Desktop Verified status	OK
Field Validated status	Partially Feasible
Slope Range in degrees	3 - 6
Distance from road	50m
Altitude	534m

Plot details and Field Observations

A seedling farm, originally located in Samtenling, which was acquired by GMC Airport, has been given substitute land of around 5 acres and currently leased. The leased land has been fenced and should be subtracted from the original area. The remaining areas of the plot are feasible.

Location Reference



Field Photos



Case 2

Plot ID	A-SP-524
Area	32.263
Location	26.894438N, 90.246166E
Desktop Verified status	OK
Field Validated status	Partially Feasible
Slope Range in degrees	4 - 9
Distance from road	150m
Altitude	633m

Plot details and Field Observations

Densely forested but the main ground is rocky terrain. Shows signs of landslides having occurred during the past. However, according to the statements of elderly locals, a landslide has never occurred during their lifetime. Not feasible for cultivation, but feasible for construction purposes. The lower portion of the plot has existing structures as well as a road running across it.

Location Reference



Field Photos



Case 3

Plot ID	A-SP-276
Area	8.698
Location	26.925187N, 90.482194E
Desktop Verified status	OK
Field Validated status	Feasible
Slope Range in degrees	14 - 16
Distance from road	Attached to farm road
Altitude	690m

Plot details and Field Observations

Feasible for cultivation and settlement, partially cultivated by Man Raj Rai and Tham Maya and family. A portion of the forested area has a steep slope and may need to be excluded from the feasible plot while some flat portions could be added into the plot.

Location Reference



Field Photos



5. THIMPHU DZONGKHAG

Case 1

Plot ID	A-TP-356
Area	1.818
Location	Kawang, Thimphu
Desktop Verified status	ok
Field Validated status	feasible
Slope Range in degrees	5-10
Distance from road	4
Altitude	2903

Plot details and Field Observations

The plot surface status is classified as partially developed/occupied, as one portion has already been developed while the remaining area, though feasible, falls under project area. This indicates a data gap, as the plot was identified as suitable during desktop analysis but was found to be under leasehold. The plot was also observed to have an access road running through it at the time of field verification.

Location Reference



Field Photos



Case 2

Plot ID	A-TP-137
Area	2.237 Acre
Location	Geney, Thimphu
Desktop Verified status	ok
Field Validated status	Feasible
Slope Range in degrees	2-10
Distance from road	2m
Altitude	2300m

Plot details and Field Observations

There is a minor spring at the upper part of the plot; otherwise, the land is suitable and located close to the access road.

Location Reference



Field Photos



Case 3

Plot ID	A-TP-443
Area	0.667
Location	Mewang,Thimphu
Desktop Verified status	ok
Field Validated status	Feasible
Slope Range in degrees	3-5
Distance from road	120m
Altitude	2523m

Plot details and Field Observations

The plot is marshy and flat, indicating it is a wetland, and it is currently used by nearby residents for storing hay for cattle.

Location Reference



Field Photos



Case 4

Plot ID	A-TP-51
Area	2.343 Acre
Location	Dagala, Thimphu
Desktop Verified status	OK
Field Validated status	Not Feasible
Slope Range in degrees	10-46
Distance from road	150m
Altitude	2692m

Plot details and Field Observations

Most parts appear marshy and steep gradient which seems unsuitable. Plot is covered by trees and bushes

Location Reference



Field Photos



6. WANGDUE PHODRANG DZONGKHAG

Case 1

Plot ID	A-WP-1267
Area	3.86
Location	Rubesa
Desktop Verified status	Ok
Field Validated status	Occupied with permanent fence and some structures
Slope Range in degrees	10-15
Distance from road	0
Altitude	1120

Plot details and Field Observations

This plot is occupied by PHPA and fully occupied with structure. Road connected till all structures; some structures are left vacant and all amenities are available.

Location Reference



Field Photo



Case 2

Plot ID	A-WP-1108
Area	8.014
Location	Athang
Desktop Verified status	Existing structure
Field Validated status	Occupied with permanent fence and some structures
Slope Range in degrees	10-20
Distance from road	0m
Altitude	780m

Plot details and Field Observations

Currently the plot is fully used as fruit tree plantation and land development.
Road connected to the plot.

Location Reference



Field Photo



Case 3

Plot ID	A-WP-1591
Area	0.718
Location	Daga
Desktop Verified status	Existing structure
Field Validated status	Occupied with permanent fence and some structures
Slope Range in degrees	0-5
Distance from road	0m
Altitude	780m

Plot details and Field Observations

Currently the plot is fully occupied with the structure of PHPA.

Location Reference



Field Photo



